# Report Gasoline

Time Series - MESIO UPC-UB

Abdelrahman Eissa & Benjamín Adasme

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## Academic Integrity Declaration

“We Abdelrahman Eissa and Benjamín Adasme hereby declare that we adhere to the UPC academic integrity commitment and that this project has been completed honestly, collaboratively, and in accordance with the established guidelines, with all sources properly acknowledged.”

# Contents

## Introduction

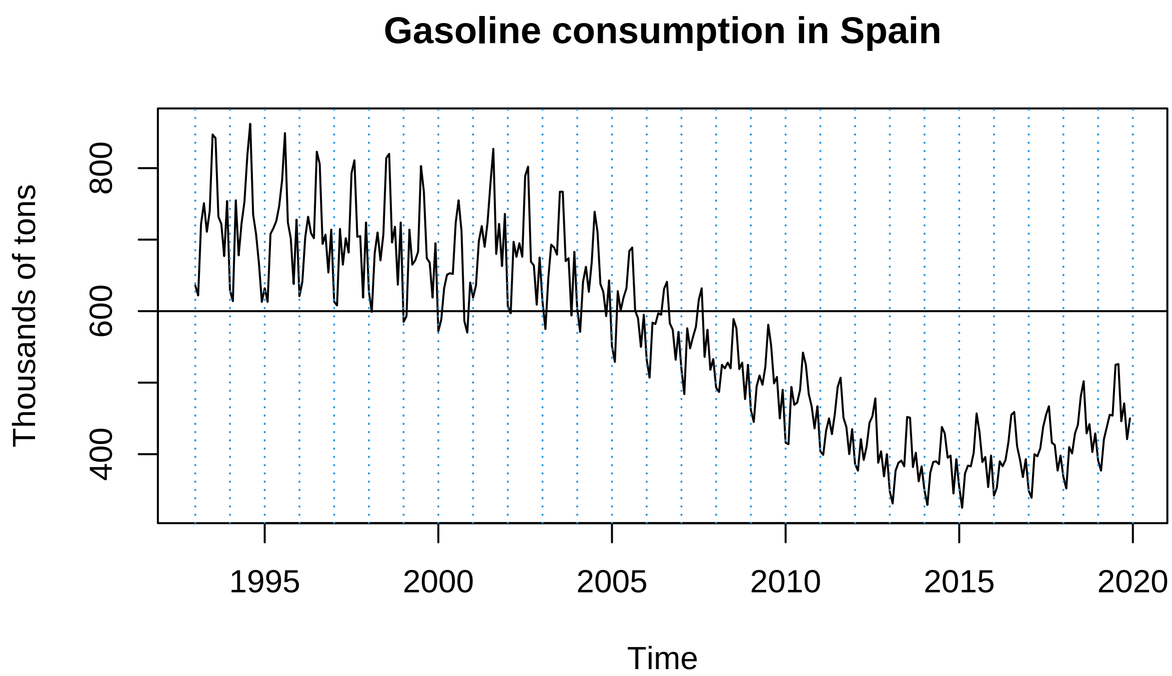
Gasoline consumption is a critical indicator of economic activity and energy demand, deeply intertwined with global and regional dynamics. Worldwide, the demand for gasoline is primarily driven by factors such as population growth, and the volatile fluctuations in global oil prices. Recently, Spanish gasoline demand has been influenced by structural shifts in the energy sector, including changing consumer habits. Furthermore, the econometric modeling and forecasting of aggregate fuel demand represent a critical area of study within energy economics and macroeconomic planning. Specifically, the analysis of gasoline consumption serves multiple strategic purposes:

- **Macroeconomic Proxy**: Liquid fuel consumption is heavily intertwined with aggregate economic activity. Because gasoline is primarily utilized for private mobility and light commercial transport, its demand acts as a highly sensitive, real-time proxy for the financial health of the national economy, reflecting fluctuations in household disposable income, consumer confidence, and overall domestic commerce.
- **The Spanish Structural Paradigm (The "Dieselization" Boom)**: Analyzing the Spanish market between 1993 and 2019 offers a unique historical case study. During the 1990s and early 2000s, Spain experienced an unprecedented structural shift toward diesel-powered vehicles—heavily incentivized by favorable national tax frameworks and European automotive manufacturing strategies. Investigating gasoline demand during this timeframe allows researchers to quantify how a traditional fuel asset contracted and adapted under intense substitution pressure from alternative fossil fuels.
- **Fiscal and Fiscal Policy Planning**: Hydrocarbon taxes constitute a substantial and predictable pillar of state revenue in Spain. Developing high-precision, short-to-medium-term forecasting models is paramount for fiscal authorities to optimize budgetary planning and accurately project tax receipts.
- **Ecological Transition and Baseline Formulation**: In the current climate policy landscape, characterized by strict European decarbonization targets and the gradual penetration of electric and hybrid drivetrains, understanding the historical, uninterrupted stochastic behavior of fossil fuel demand is vital. This study establishes a rigorous "baseline" of traditional consumption patterns, which is essential for evaluating the real-world impact of ongoing energy transition policies.

To analyze and forecast these dynamics, this study utilizes the "Dataset Gasoline." This dataset provides a comprehensive time series of historical gasoline consumption in Spain. By capturing both the long-term trends and the short-term seasonal or anomalous shocks such as sudden price hikes or economic downturns, the dataset offers a robust foundation for modeling the underlying behavior of Spanish fuel demand.

### Objective

The report focuses on modeling and forecasting gasoline consumption in Spain. To accurately capture the trends and variations in the data, the study contrasts two primary time series forecasting methodologies: a classic Box-Jenkins ARIMA approach and an advanced ARIMAX approach. The core objective is to identify a statistically valid model that can reliably predict future gasoline consumption by correctly handling underlying patterns, external calendar variations, and unexpected shocks to the dataset.




### Methodology

This study follows a rigorous and structured approach to time series forecasting, primarily divided into two main analytical frameworks: the Classic ARIMA approach and the ARIMAX approach. The goal is to systematically identify the underlying patterns in the gasoline consumption dataset, account for external variations, and develop a statistically robust model for accurate forecasting.

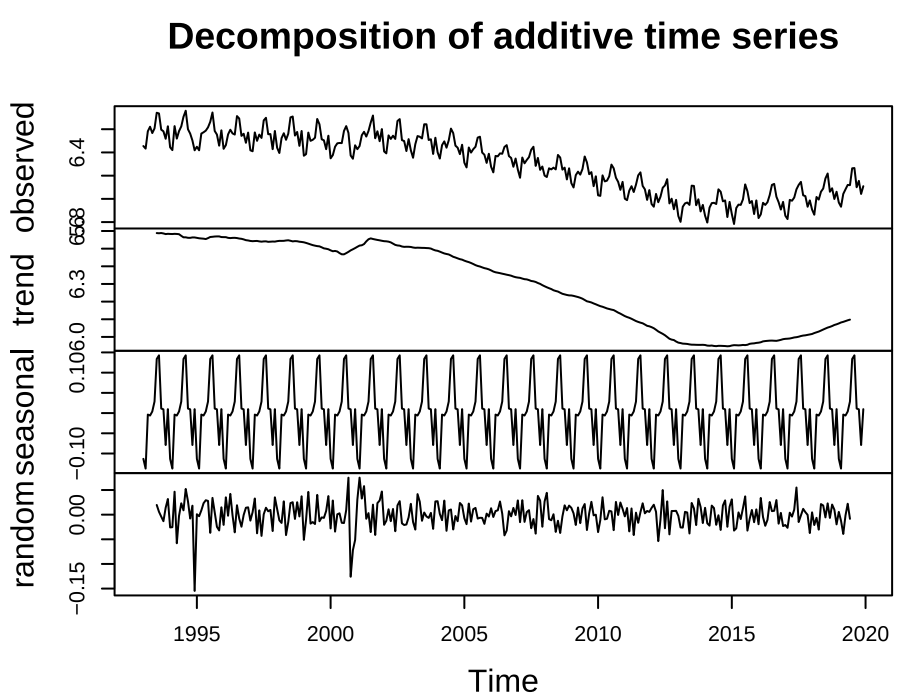



Figure 1: Decomposition of log-transformed series

**Variance and Mean Stabilization (Stationarity)**

A fundamental prerequisite for Box-Jenkins modeling is that the underlying stochastic process must be stationary in both variance and mean.

- Variance Stabilization: Prior to any linear differentiation, the original gasoline consumption series $\{Y_t\}$ was evaluated to detect heteroskedasticity. Since the amplitude of the seasonal oscillations changed proportionally with the long-term trend, a natural logarithm transformation was applied to stabilize the conditional variance:

$$Z_t = \ln(Y_t)$$

This logarithmic transformation linearizes the exponential components and ensures that the variance remains constant ($\sigma^2$) across the entire historical horizon.

- Mean Stabilization (Differentiation): To eliminate non-stationarity in the mean caused by the secular downward trend and the cyclical annual patterns, log-differencing operators were implemented. First, a regular first-difference operator ($\nabla = 1 - B$) was applied to remove the trend. Second, a seasonal difference operator ($\nabla_{12} = 1 - B^{12}$) with a period of $s = 12$ was applied to eliminate the annual seasonal component, where $B$ represents the backshift (lag) operator ($B^k X_t = X_{t-k}$). The fully stationarized series $W_t$ is defined as:

$$W_t = \nabla\nabla_{12} Z_t = (1 - B)(1 - B^{12})\ln(Y_t)$$

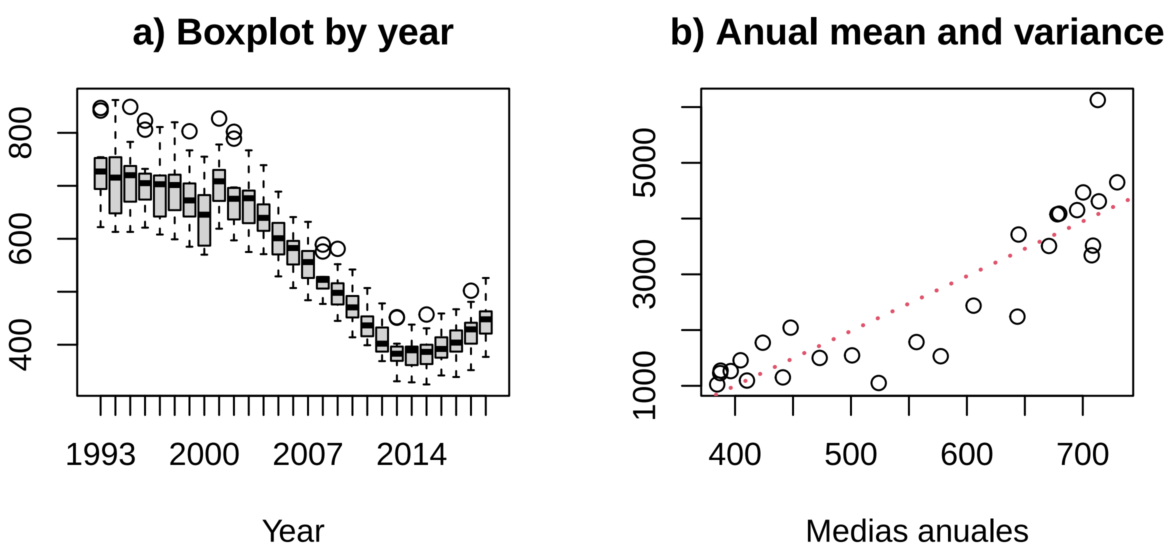



Figure 2: Non-constant variance series

**Classic ARIMA Approach**

For the analysis, we followed the iterative Box-Jenkins methodology through its stages of identification, estimation, validation, and prediction. To transform the non-stationary series into a stationary one, we first stabilized the variance using a logarithmic transformation, and subsequently corrected the non-stationarity in the mean by applying both regular and seasonal differencing. The methodological design is further extended through the modeling of calendar effects—accounting for working days and the moving Easter effect—as well as the automatic detection of additive outliers, transitory changes, and level shifts. -Identification: Once stationarity is achieved, the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) are plotted and analyzed to identify potential autoregressive (AR) and moving average (MA) terms, yielding a set of plausible candidate models. - Estimation: For each of the proposed candidate models, the parameters (coefficients for the AR and MA terms) are estimated, typically using Maximum Likelihood Estimation (MLE). During this phase, the models are evaluated and compared using goodness-of-fit criteria, primarily

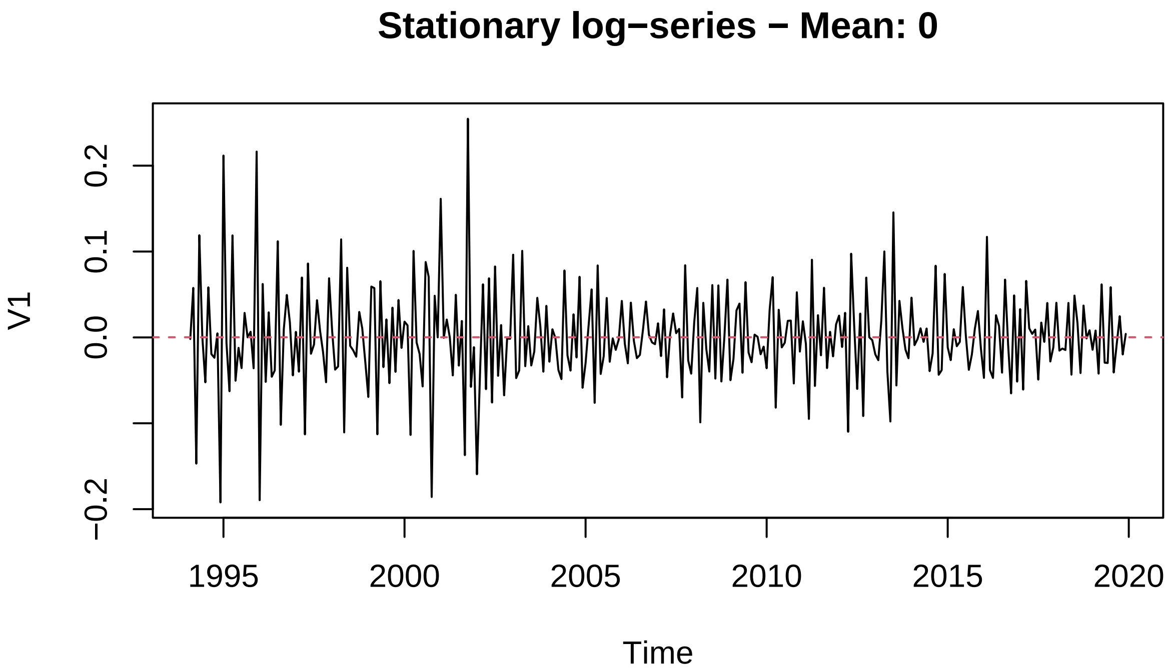



Figure 3: Stationary series

the Akaike Information Criterion (AIC), to determine which model best captures the data's structure with the optimal number of parameters. - Diagnosis and validation: The selected models are then subjected to rigorous diagnostic checks to ensure their statistical validity. This step focuses on analyzing the models' residuals, which should ideally behave as white noise. Various statistical tests are employed, including the Ljung-Box test to verify the absence of autocorrelation, the Shapiro-Wilk or Jarque-Bera tests for normality, and tests for homoscedasticity. - Prediction and forecasting: Once a valid model is established, it is used to generate point forecasts and confidence intervals for future gasoline consumption. The model's predictive accuracy is evaluated by comparing out-of-sample forecasts against actual data using standard error metrics such as the Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), confidence interval and other amtrics.

### ARIMAX Approach

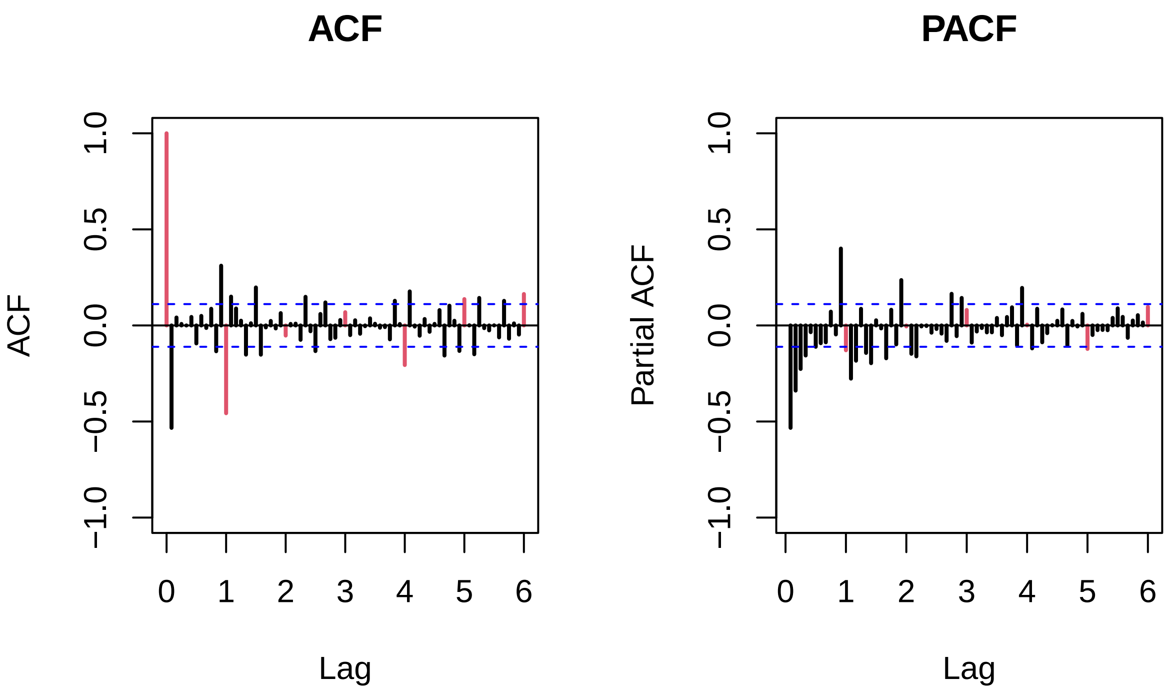
Since real-world economic data like gasoline consumption is heavily influenced by exogenous events and calendar anomalies, the classic ARIMA framework is often insufficient on its own. The second phase of the methodology addresses these limitations through an ARIMAX. - Calendar Effects: We introduce dummy variables to account for known calendar variations that systematically impact fuel consumption, such as the trading day effect (the composition of weekdays versus weekends in a given month) and the Easter effect. - Outlier Treatment: An automatic outlier detection procedure is applied to identify significant anomalies in the historical data, such as Additive Outliers (AO), Level Shifts (LS), and Temporary Changes (TC). These outliers can distort parameter estimation and residual diagnostics if left untreated. - Arimax Remodeling: The original time series is "cleaned" by adjusting for the identified calendar effects and deterministic outliers. With this adjusted series, the entire Box-Jenkins process (Identification, Estimation, Diagnosis, and Prediction) is repeated. This results in an ARIMAX model that inherently controls for past anomalies and external factors, theoretically yielding a more stationary residual series and more precise, reliable forecasts.

## Results and discussion

## Classic ARIMA approach

### Estimation for plausible models

In the model identification phase of the Box-Jenkins methodology, we analyze the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots of the suitably differenced stationary time series to determine the appropriate autoregressive (AR) and moving average (MA) terms. Based on the visual inspection of the provided correlograms, we divide our identification process into two parts: the seasonal component and the regular (non-seasonal) component.




**Seasonal Component Identification**: Because PACF shows significant activity at the first seasonal lag—rather than a clear, sharp cutoff that is shown in the ACF at the first seasonal lag moreover, a mixed seasonal process is the most appropriate choice. Therefore, we select a Seasonal ARMA(1,1) model to capture both the seasonal autoregressive (SAR) and seasonal moving average (SMA).

**Regular Component Identification**:

- MA(1): Proposed under the assumption that the regular ACF effectively cuts off after the very first lag, while the PACF tails off
- ARMA(1,1): assuming both the ACF and PACF exhibit a tapering/decaying pattern from the first lag, capturing both a short-term autoregressive and moving average effect.
- MA(2): Proposed to account for the possibility that the first two lags in the ACF are driving the moving average process before cutting off
- ARMA(1,2) : mixed model included to capture any residual moving average dependencies up to the second lag while maintaining a first-order autoregressive term.

As shown in the table above, an estimation comparison was conducted between the 4 selected models based on their goodness of fit and complexity. The results of **log likelihood** and **AIC** values indicate that **MA(2)** and **ARMA(1,2)** exhibit the two best overall performing models.

Both models' residuals were evaluated, reaching the normality and independence requirements, as can be see in the next figures. However, the Ljung-Box test shows significant p-values starting lag 12 approximately,

Table 1: Model Comparison

| Models | Log-Likelihood | AIC | AR(1) | MA(1) | MA(2) | SMA(1) |
|---|---|---|---|---|---|---|
| MA(1) | 610.97 | -1215.93 | NA | -0.68 | NA | -0.89 |
| ARMA(1,1) | 613.46 | -1218.93 | -0.17 | -0.60 | NA | -0.88 |
| **MA(2)** | **613.95** | **-1219.89** | **NA** | **-0.79** | **0.14** | **-0.88** |
| **ARMA(1,2)** | **617.75** | **-1225.50** | **0.99** | **-1.73** | **0.75** | **-0.92** |

Table 2: MA(2)(left) and ARMA(1,2)(right) stability

| Variables | Full | Restricted | Diference |
|---|---|---|---|
| ma1 | -0.79 | -0.78 | 0.01 |
| ma2 | 0.14 | 0.13 | 0.01 |
| sma1 | -0.88 | -0.89 | 0.01 |

| Variables | Full | Restricted | Diference |
|---|---|---|---|
| ar1 | 0.99 | 0.99 | 0.00 |
| ma1 | -1.73 | -1.72 | 0.01 |
| ma2 | 0.75 | 0.74 | 0.01 |
| sma1 | -0.92 | -0.93 | 0.01 |

which is problematic since it means certain degree of autocorrelation in residuals[1].

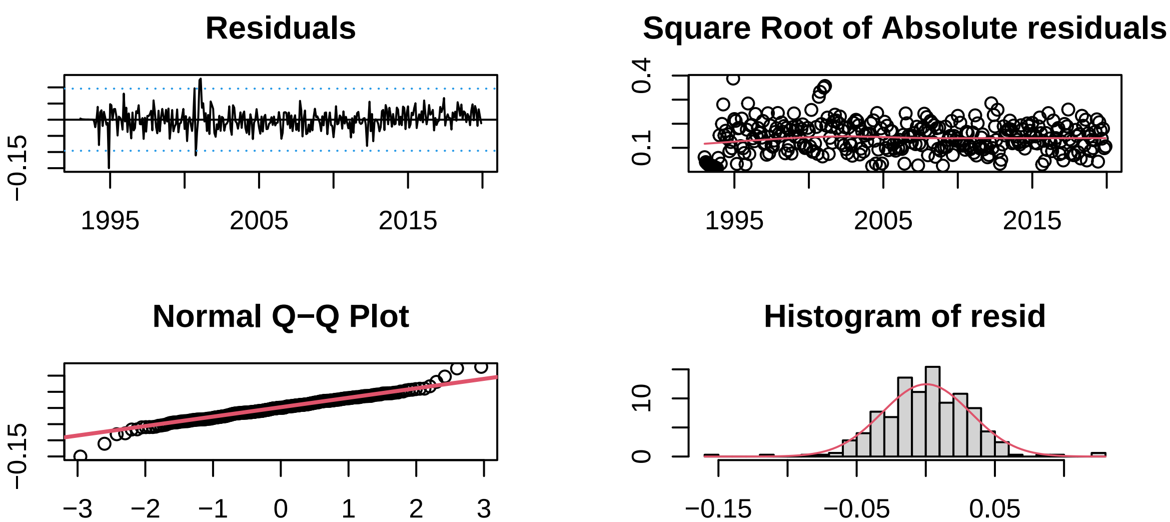



Figure 4: MA(2) residual validation

**Prediction assesment and forecasting**

To evaluate the prediction performance, we checked the models' stability, it means the comparison between the coefficients of the model with the full series and with a restricted series without the las 12 observations. Both MA(2) and ARMA(1,2) showed stability since their coefficients don't change significantly under those conditions.

In terms of prediction, the plots show how MA(2) and ARMA(1,2) perform trying to forecast the values for the last year. Visualy speaking, we can see ARMA(1,2) has predicted points (in red) closer to the real value points (in black).

The metrics also show better performance for ARMA(1,2) mainly for RMSE (12.4) and MAE (9.39). Both models reach a mean confidence interval close to 63, and very similar RMSPE and MAPE.

[1] See details in Appendices

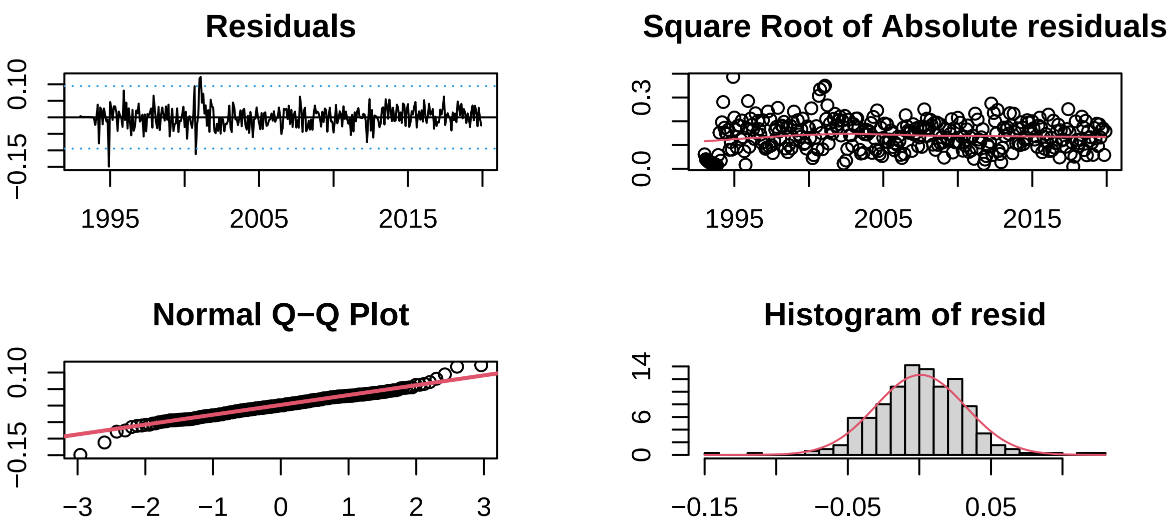



Figure 5: ARMA(1,2) residual validation

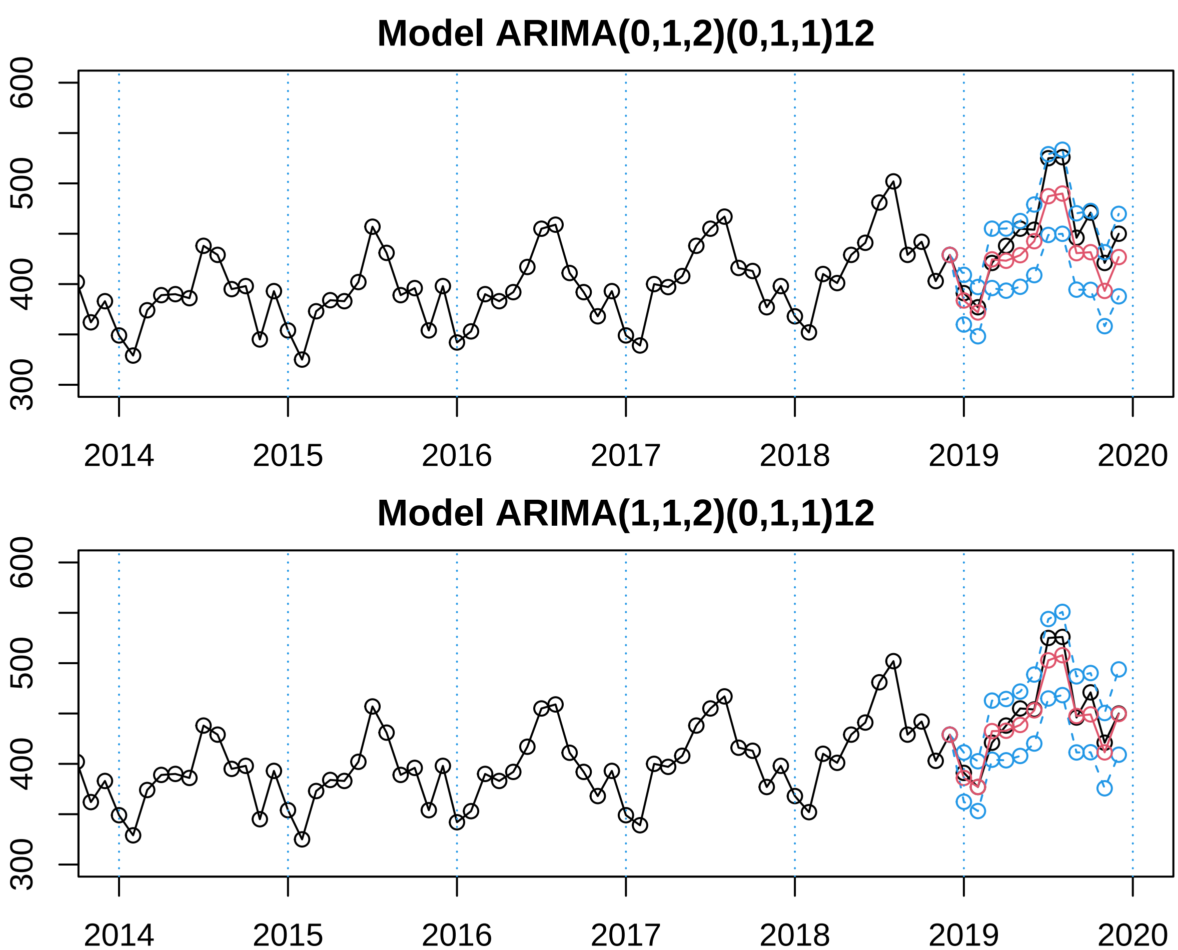



Figure 6: Prediction assesment for the last 12 months

| Model | RMSE | MAE | RMSPE | MAPE | Mean CI |
|---|---|---|---|---|---|
| ARIMA(0,1,2) | 24.06 | 20.66 | 0.05 | 0.04 | 63.65 |
| ARiMA(1,1,2) | 12.40 | 9.39 | 0.03 | 0.02 | 63.52 |

At this point, our best model by Information Criteria and prediction performance seems to be the SARIMA(1,1,2)x(0,1,1), which can be formally described as:

$$(1-\phi_1 B)(1-B)(1-B^{12})X_t = (1+\theta_1 B+\theta_2 B^2)(1+\Theta_1 B^{12})Z_t$$
$$(1-0.99B)(1-B)(1-B^{12})_t = (1-1.73B+0.75B^2)(1+-0.92B^{12})Z_t$$

With this full model, the forecast for the next 12 months is described in the next plot. As can be expected, the prediction follows the seasonal pattern very well, imitating the anual shape of the curve.

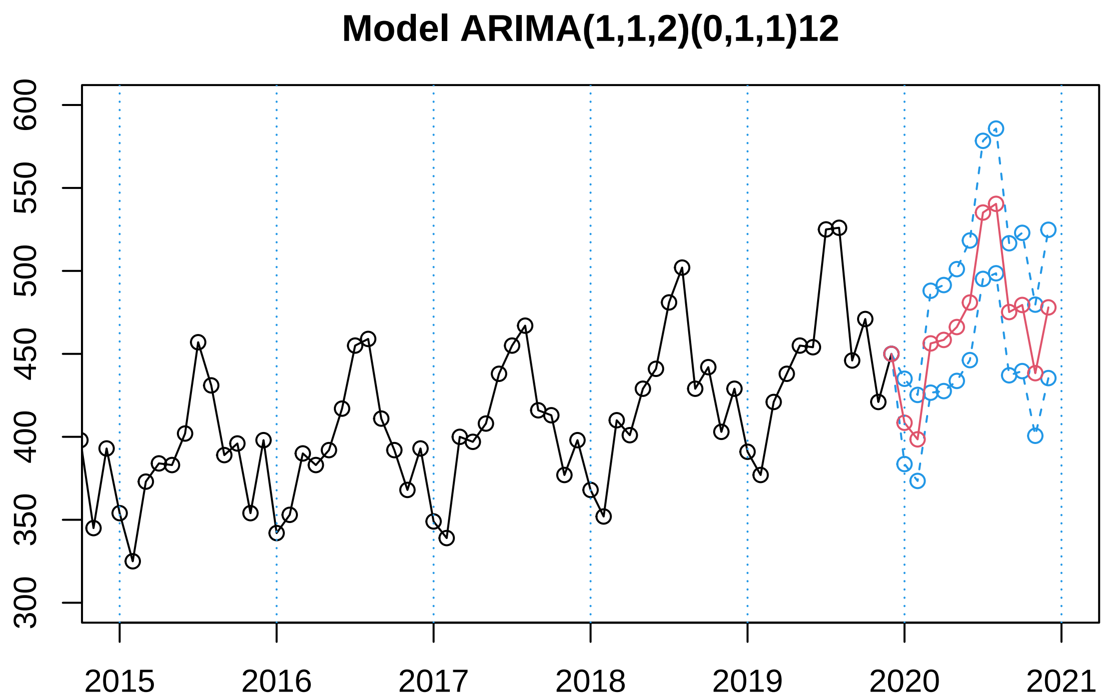



Figure 7: SARIMA(1,1,2)x(0,1,1)12

However, the selected model still have a big problem: auto-regresive residuals. As we mentioned before, none of the proposed models passed the Ljung-Box test, which means that there are certain degree of auto-correlation in the residuals of the models. Since none of the proposed model avoid the problem we follow the next hypothesis: this could be corrected by taking into account some calendar effects and with automatic outlier detection.

#### ARIMAX Analysis: Outlier and Calendar Effects Treatment

Following our hypothesis, and according to the nature of gasoline consumption, we belive there are two main calendar effects that are affecting our series: labor days and Easter holidays. It follows because on labor days there is more transit due to labor and educational activities. This pushes up the gasoline consumption and with more labor days in a month the consumption will rise. Easter holidays are a special case since it implies a high level of tourism and travels, but also because it changes its dates every year between March and April. So, we should take it into account these two conditions as important calendar effects.

To test the relevance of calendar effects we are creating two dummy variables to identify labor days and Easter across our series. Next, we use those as exogenous variables in our best model (in this case, ARIMA(1,1,2)). Then we test the significance of the model's terms as the ratio between coefficients and standard errors.

Table 4: Calendar Effect: coefficient validation

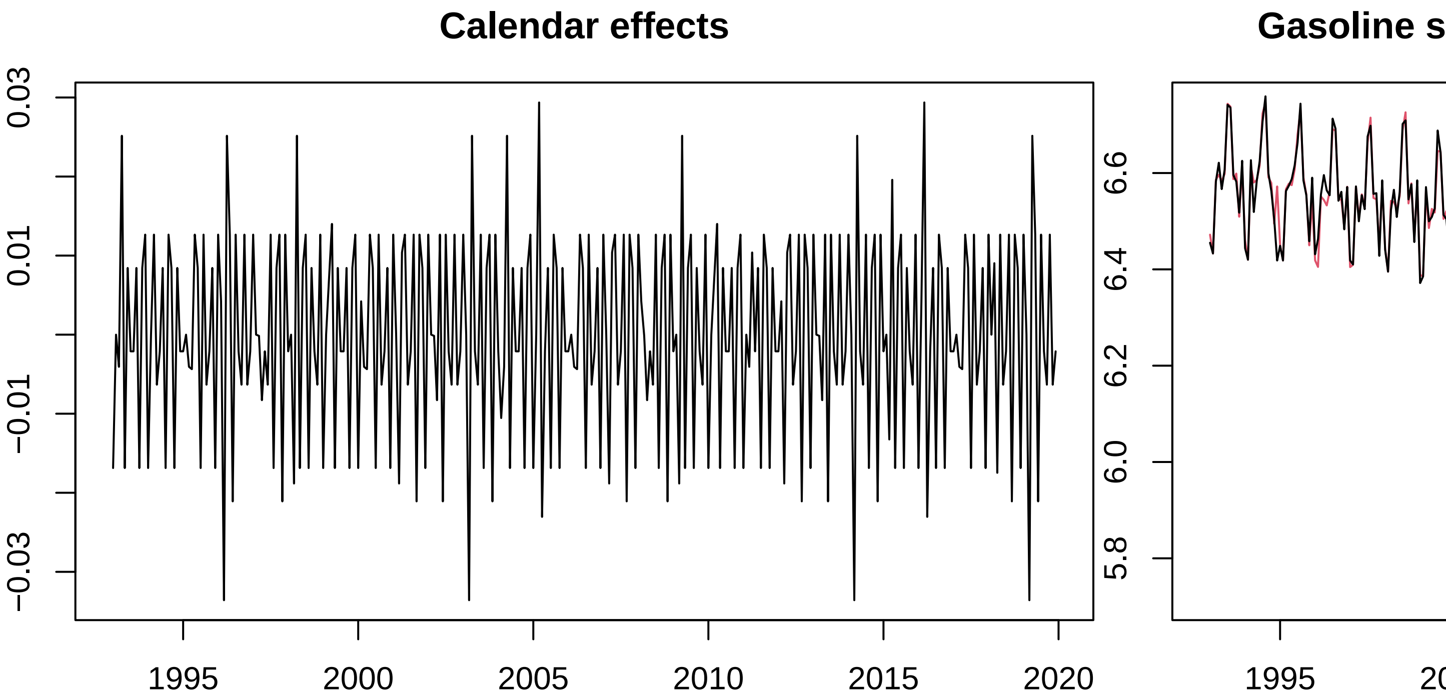
| term | estimate | std.error | Significance |
|---|---|---|---|
| ar1 | 0.99 | 0.01 | 66.25 |
| ma1 | -1.68 | 0.04 | -39.62 |
| ma2 | 0.70 | 0.04 | 17.13 |
| sma1 | -0.90 | 0.04 | -21.52 |
| wTradDays | 0.00 | 0.00 | 8.11 |
| wEast | 0.03 | 0.01 | 3.94 |

As can be seen in the table, the exogenous variables show to be significant to explain the observed series. After this, we are using the `outdetec` function to detect different kind of outliers: additive outliers, transitive change and level shift. The next table shows the observations identified as outlier and their classification.

Table 5: Detected outliers in Gasoline consumption series

| | Obs | type_detected | W_coeff | ABS_L_Ratio | Date | perc.Obs |
|---|---|---|---|---|---|---|
| 10 | 16 | AO | -0.055 | 2.968 | Abr 1994 | -0.055 |
| 2 | 24 | AO | -0.152 | 6.876 | Dic 1994 | -0.152 |
| 7 | 36 | AO | 0.064 | 3.313 | Dic 1995 | 0.064 |
| 9 | 38 | TC | 0.054 | 3.171 | Feb 1996 | 0.054 |
| 19 | 79 | AO | 0.045 | 2.674 | Jul 1999 | 0.045 |
| 3 | 93 | LS | 0.092 | 5.596 | Sep 2000 | 0.092 |
| 1 | 94 | TC | -0.143 | 6.650 | Oct 2000 | -0.143 |
| 4 | 97 | LS | 0.069 | 4.325 | Ene 2001 | 0.069 |
| 15 | 98 | AO | 0.046 | 2.641 | Feb 2001 | 0.046 |
| 8 | 108 | AO | 0.064 | 3.366 | Dic 2001 | 0.064 |
| 14 | 164 | AO | -0.047 | 2.662 | Ago 2006 | -0.047 |
| 11 | 171 | AO | 0.049 | 2.694 | Mar 2007 | 0.049 |
| 18 | 182 | AO | 0.046 | 2.691 | Feb 2008 | 0.046 |
| 12 | 186 | TC | -0.045 | 2.723 | Jun 2008 | -0.045 |
| 16 | 219 | LS | -0.036 | 2.681 | Mar 2011 | -0.036 |
| 5 | 232 | LS | -0.054 | 3.428 | Abr 2012 | -0.054 |
| 6 | 234 | AO | 0.081 | 4.106 | Jun 2012 | 0.081 |
| 17 | 278 | AO | 0.046 | 2.631 | Feb 2016 | 0.046 |
| 13 | 294 | AO | 0.048 | 2.632 | Jun 2017 | 0.048 |

At this point, with calendar effects and outlier detection, we can "clean" our series taking out those atypical patterns. The new series will be a linear combination of the original one and its outliers. Then, we will fit our selected model over this new series, with calendar effects as regressors to extract their coefficients. With estimated effects, our clean series would result from the free-of-outlier series minus their calendar effects. Here we can see the estimated calendar effects and our original series along with the clean one.




**ARIMAX model identification**

Once we have our new series, we repeat the identification process from the ACF and PACF plots. For this series we are considering again one regular and one seasonal differentiation to keep it stationary.

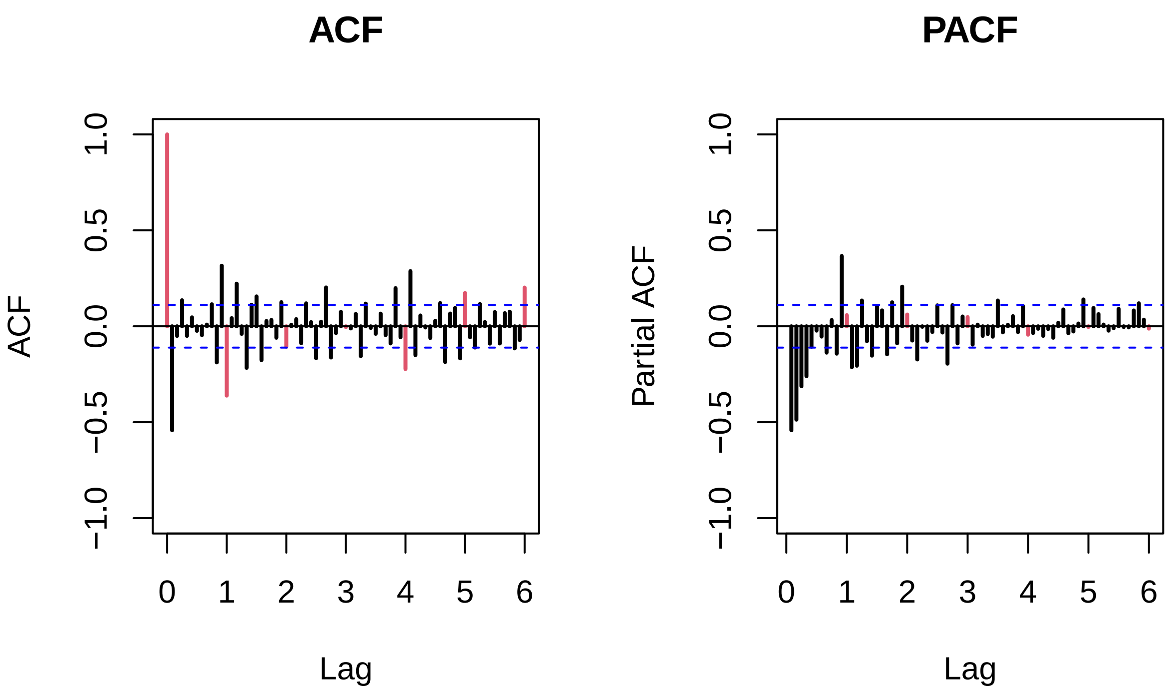



Figure 8: ARIMAX model identification

Proposed models for clean series: Regular MA(1), MA(3), MA(11) and ARIMA(1,2) with Seasonal MA(1).

The first insight we extract from the plots is the clear seasonal pattern, which can be described as an moving average of level 1 (MA(1)). For the regular part we identify this possible combinations: MA(1), MA(3), MA(11), ARMA(1,5) and ARMA(1,2)[2]. After fitting and testing the candidates, we compare them by Log-likelihood and AIC.

Table 6: ARIMAX Models Comparison

| ARIMAX Model | Log-Likelihood | AIC |
|---|---|---|
| MA(1) | 775.46 | -1540.93 |
| MA(2) | 777.30 | -1542.61 |
| **MA(11)** | **795.44** | **-1560.89** |
| **ARMA(1,2)** | **789.87** | **-1565.74** |
| ARMA(1,5) | 782.57 | -1545.13 |

In general terms, the five models reach an important goal: they improve the AIC of ARIMA models by far, from around -1200 to -1500. The two best performing models were MA(11) and ARMA(1,2). The first one is an model adjusted using "brute force", which works but makes the model much more complex. On the other hand, ARMA(1,2) keeps the structure identified in the ARIMA approach, and here performs very well reaching the best AIC. Also both of them achieve all validation requirements, specially the Ljung-Box test that failed for the ARIMA models. For these reasons, we will test their performance in prediction, using the last year as the test sample.

Table 7: ARIMAX Models Prediction assessment

| Model | RMSE | MAE | RMSPE | MAPE | Mean CI |
|---|---|---|---|---|---|
| ARIMAX(0,1,11) | 34.71 | 35.37 | 0.08 | 0.08 | 34.63 |
| ARIMAX(1,1,2) | 30.36 | 30.72 | 0.07 | 0.07 | 35.36 |

**ARIMA and ARIMAX models comparison**

Finally, we proceed to compare the two ARIMA and the two ARIMAX models. Here are our results.

Table 8: ARIMA and ARIMAX models comparison

| Models | Log-Likelihood | AIC | RMSE | MAE | RMSPE | MAPE | Mean CI |
|---|---|---|---|---|---|---|---|
| ARIMA(0,1,2) | 613.95 | -1219.89 | 24.06 | 20.66 | 0.05 | 0.04 | 63.65 |
| ARiMA(1,1,2) | 617.75 | -1225.50 | 12.40 | 9.39 | 0.03 | 0.02 | 63.52 |
| ARIMAX(0,1,11) | 795.44 | -1560.89 | 34.71 | 35.37 | 0.08 | 0.08 | 34.63 |
| **ARIMAX(1,1,2)** | **789.87** | **-1565.74** | **30.36** | **30.72** | **0.07** | **0.07** | **35.36** |

Looking the results, we realized there is a paradox. On one side we have ARIMA models with great point prediction metrics. We could think this is perfect for forecast and future decisions about the gasoline. But they don't take into account the outilers and calendar effects. On the other side, ARIMAX models don't perform as well in RMSE, MAE and other metrics. Instead, they improve significantly the AIC score and reduce by almost half the mean confidence interval. In other words, ARIMA models provide good prediction but with high uncertainty while ARIMAX models don't are too pointly accurate but they are less uncertain and provide best information criteria. Which on is the better choice?

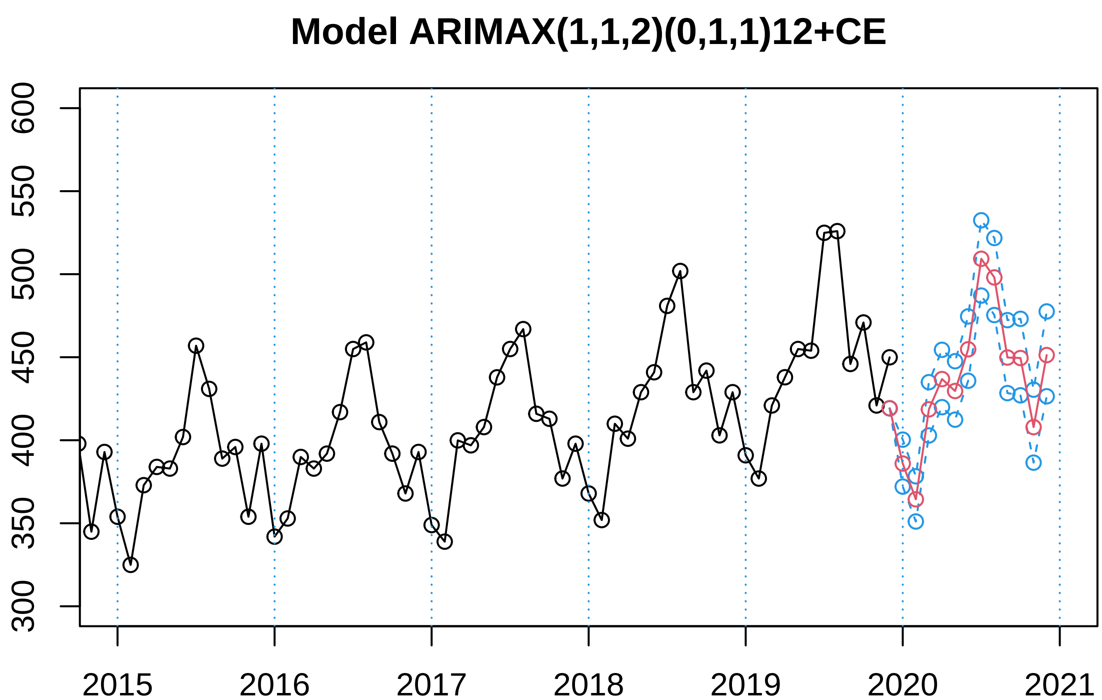
[2]ARMA(1,2) was tested because it was our selected model on ARIMA approach

We suggest that the best model for the gasoline consumption is **ARIMAX(1,1,2)**. Despite it may look a weak model for prediction, we find several reasons to choose it. The empirical findings of this study demonstrate that evaluating time series models solely through traditional point-prediction error metrics—such as RMSE or MAE—can be fundamentally misleading. While an uncorrected ARIMA model may occasionally yield lower localized errors during a specific test horizon, this phenomenon is often an artifact of overfitting to historical noise rather than a reflection of true predictive power. In contrast, the integration of deterministic components via the SARIMAX framework proved statistically essential. By simultaneously controlling for holiday variations, trading day variations, and historical anomalies through the incorporation of calendar effects and outliers, the underlying stochastic signal of the series was successfully isolated.

This structural refinement resulted in a dramatic improvement in the Akaike Information Criterion (AIC), falling from -1225.50 in the univariate ARIMA model to -1565.74 in the optimal ARIMAX specification. Furthermore, by accounting for these external shocks within the exogenous regressor matrix (xreg), the model achieved a high degree of parsimony. It effectively substituted a convoluted, overparameterized error structure for an elegant ARIMAX$(1,1,2)$ specification that relies on far fewer parameters while ensuring that the model's residuals successfully conform to white noise under the Ljung-Box test. Most importantly for operational decision-making, extracting this systemic variance drastically reduced forecasting uncertainty, shrinking the mean prediction interval width nearly in half (from 63.52 to 35.36). Consequently, the ARIMAX$(1,1,2)$ model stands as the mathematically superior and most risk-averse framework for long-term strategic planning, capable of dynamically adapting to future calendar structures without being derailed by past anomalous volatility.

The final model can be written in it formula as:

$$(1-0.9703B)(1-B)(1-B^{12})\,[y_t - 0.0042\cdot\mathrm{wTradDays}_t - 0.0334\cdot\mathrm{wEast}_t] = (1-1.7674B+0.8044B^2)(1-0.8272B^{12})Z_t$$




## Conclusions

The empirical findings of this study highlight a critical econometric paradox: relying solely on localized point-prediction errors (such as RMSE or MAE) can be highly misleading. While an uncorrected ARIMA model may occasionally mimic the data quite closely over a specific test horizon, this phenomenon is typically driven by sheer luck or severe overfitting to historical noise. Because it leaves systematic patterns unaddressed, the univariate model fails basic residual diagnostics, rendering its forecasts statistically invalid.

In contrast, the ARIMAX advantage lies in its ability to simultaneously isolate structural shocks from the true underlying stochastic process. By integrating calendar effects and historical outliers directly into the model via a SARIMAX$(1, 1, 2)$ specification, we successfully capture the series' true dynamics, eliminate persistent residual patterns, and comfortably pass the Ljung-Box test for white noise.

Crucially, accounting for these external variations through simultaneous exogenous regressors (xreg) drastically reduces forecasting uncertainty. By absorbing the volatile "swings" caused by trading days and holidays, the model slashes the mean prediction interval width nearly in half. This provides decision-makers with a highly tightened, risk-managed boundary for future planning rather than a wide, uninformative margin of error.

Ultimately, this framework ensures a highly strategic projection into the future. By relying on a parsimonious yet robust mathematical structure, the model is fully equipped to seamlessly adapt to upcoming calendar configurations for unobserved years while safely neutralizing the echo of past anomalous shocks.

## Appendices

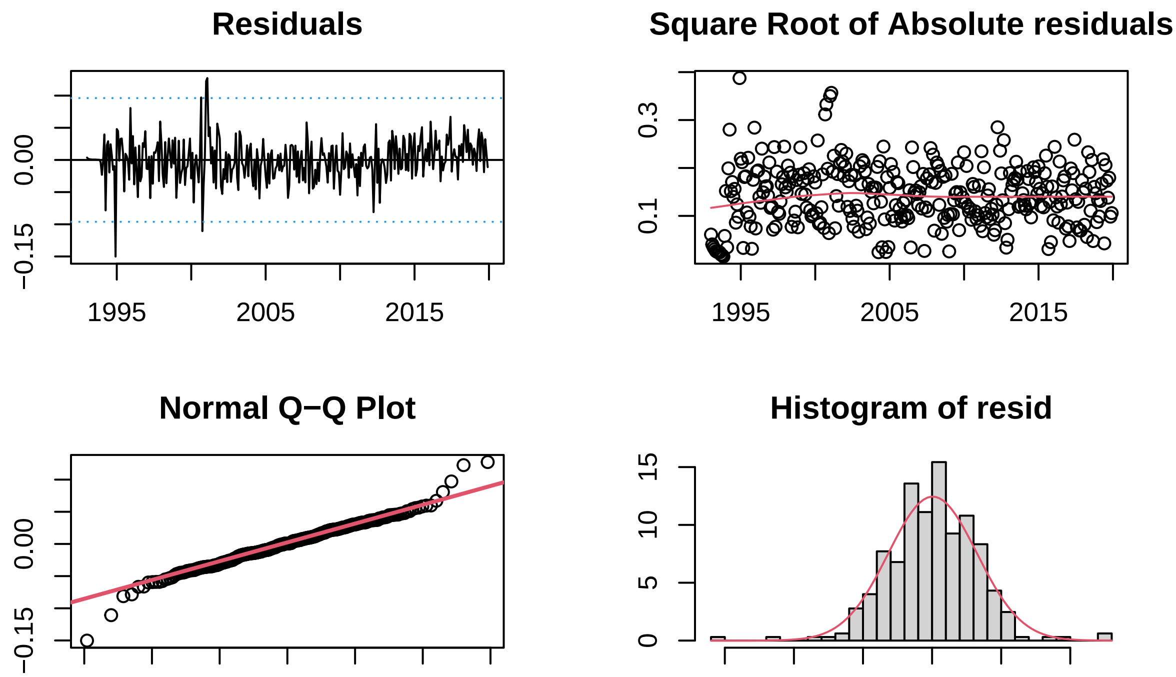
### Validation function

```r
################Validation#################################
validation=function(model,dades){
  s=frequency(get(model$series))
  resid=model$residuals
  par(mfrow=c(2,2),mar=c(3,3,3,3))
  #Residuals plot
  plot(resid,main="Residuals")
  abline(h=0)
  abline(h=c(-3*sd(resid),3*sd(resid)),lty=3,col=4)
  #Square Root of absolute values of residuals (Homocedasticity)
  scatter.smooth(sqrt(abs(resid)),main="Square Root of Absolute residuals",
               lpars=list(col=2))

  #Normal plot of residuals
  qqnorm(resid)
  qqline(resid,col=2,lwd=2)

  ##Histogram of residuals with normal curve
  hist(resid,breaks=20,freq=FALSE)
  curve(dnorm(x,mean=mean(resid),sd=sd(resid)),col=2,add=T)


  #ACF & PACF of residuals
  par(mfrow=c(1,2))
  acf(resid,ylim=c(-1,1),lag.max=60,col=c(2,rep(1,s-1)),lwd=1)
  pacf(resid,ylim=c(-1,1),lag.max=60,col=c(rep(1,s-1),2),lwd=1)
  par(mfrow=c(1,1))

  #ACF & PACF of square residuals
  par(mfrow=c(1,2))
  acf(resid^2,ylim=c(-1,1),lag.max=60,col=c(2,rep(1,s-1)),lwd=1)
  pacf(resid^2,ylim=c(-1,1),lag.max=60,col=c(rep(1,s-1),2),lwd=1)
  par(mfrow=c(1,1))

  #Ljung-Box p-values
  par(mar=c(2,2,1,1))
  tsdiag(model,gof.lag=7*s)
  cat("\n--------------------------------------------------------------------\n")
  print(model)

  #Stationary and Invertible
  cat("\nModul of AR Characteristic polynomial Roots: ",
      Mod(polyroot(c(1,-model$model$phi))),"\n")
  cat("\nModul of MA Characteristic polynomial Roots: ",
      Mod(polyroot(c(1,model$model$theta))),"\n")

  #Model expressed as an MA infinity (psi-weights)
  psis=ARMAtoMA(ar=model$model$phi,ma=model$model$theta,lag.max=36)
  names(psis)=paste("psi",1:36)
```

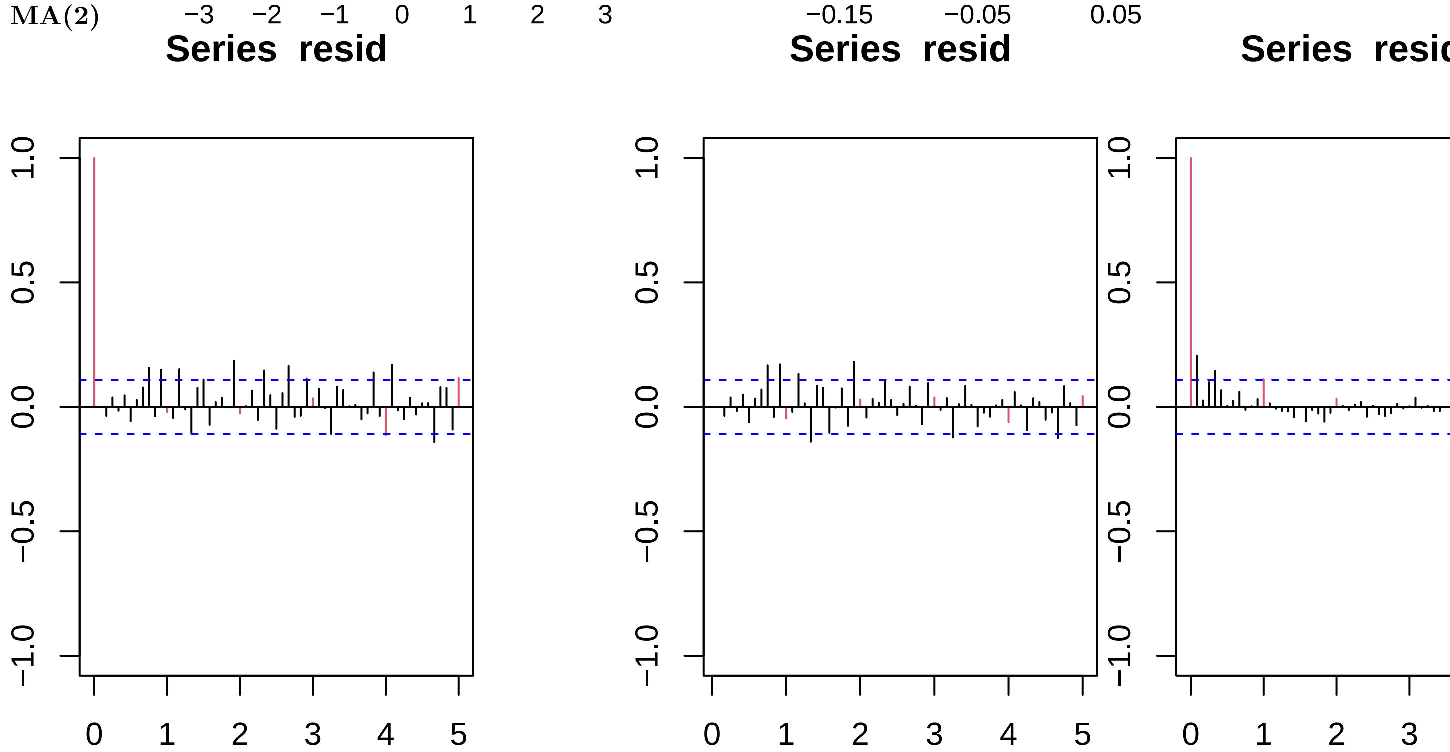
```r
cat("\nPsi-weights (MA(inf))\n")
cat("\n--------------------\n")
print(psis[1:20])

#Model expressed as an AR infinity (pi-weights)
pis=-ARMAtoMA(ar=-model$model$theta,ma=-model$model$phi,lag.max=36)
names(pis)=paste("pi",1:36)
cat("\nPi-weights (AR(inf))\n")
cat("\n--------------------\n")
print(pis[1:20])

## Add here complementary tests (use with caution!)
##---------------------------------------------------------
cat("\nNormality Tests\n")
cat("\n--------------------\n")

##Shapiro-Wilks Normality test
print(shapiro.test(resid(model)))

suppressMessages(require(nortest,quietly=TRUE,warn.conflicts=FALSE))
##Anderson-Darling test
print(ad.test(resid(model)))

suppressMessages(require(tseries,quietly=TRUE,warn.conflicts=FALSE))
##Jarque-Bera test
print(jarque.bera.test(resid(model)))

cat("\nHomoscedasticity Test\n")
cat("\n--------------------\n")
suppressMessages(require(lmtest,quietly=TRUE,warn.conflicts=FALSE))
##Breusch-Pagan test
obs=get(model$series)
print(bptest(resid(model)~I(obs-resid(model))))

cat("\nIndependence Tests\n")
cat("\n--------------------\n")

##Durbin-Watson test
print(dwtest(resid(model)~I(1:length(resid(model)))))

##Ljung-Box test
cat("\nLjung-Box test\n")
print(t(apply(matrix(c(1:4,(1:4)*s)),1,function(el) {
  te=Box.test(resid(model),type="Ljung-Box",lag=el)
  c(lag=(te$parameter),statistic=te$statistic[[1]],p.value=te$p.value)})))


#Sample ACF vs. Teoric ACF
par(mfrow=c(2,2),mar=c(3,3,3,3))
acf(dades, ylim=c(-1,1) ,lag.max=36,main="Sample ACF")

plot(ARMAacf(model$model$phi,model$model$theta,lag.max=36),ylim=c(-1,1),
     type="h",xlab="Lag",  ylab="", main="ACF Teoric")
```

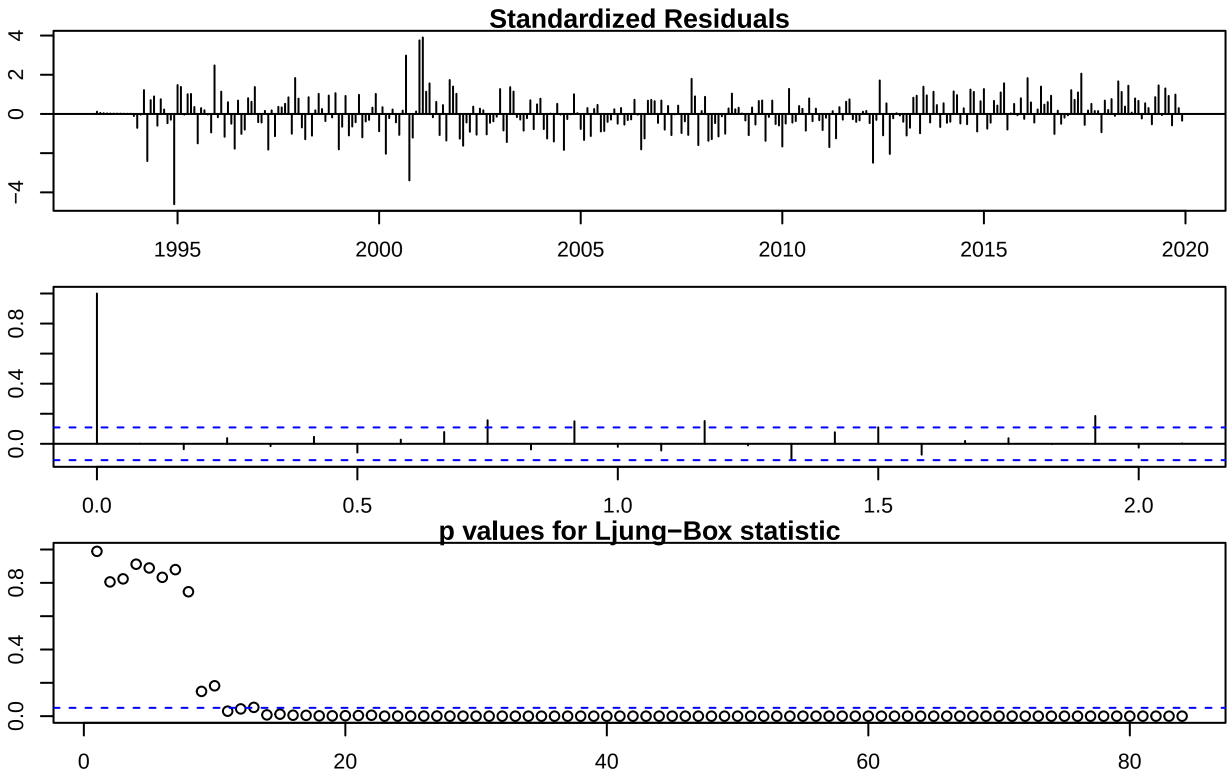
```
  abline(h=0)

  #Sample PACF vs. Teoric PACF
  pacf(dades, ylim=c(-1,1) ,lag.max=36,main="Sample PACF")

  plot(ARMAacf(model$model$phi,model$model$theta,lag.max=36, pacf=T),ylim=c(-1,1),
       type="h", xlab="Lag", ylab="", main="PACF Teoric")
  abline(h=0)
  par(mfrow=c(1,1))
}
################# Fi Validation ##################################
```

**ARIMA models validation detailed**


Residuals
−0.15
0.00
1995
2005
2015
Square Root of Absolute residuals
0.1
0.3
1995
2005
2015
Normal Q−Q Plot
−0.15
0.00
−3
−2
−1
0
1
2
3
Histogram of resid
0
5
10
15
−0.15
−0.05
0.05

$\mathbf{MA(2)}$

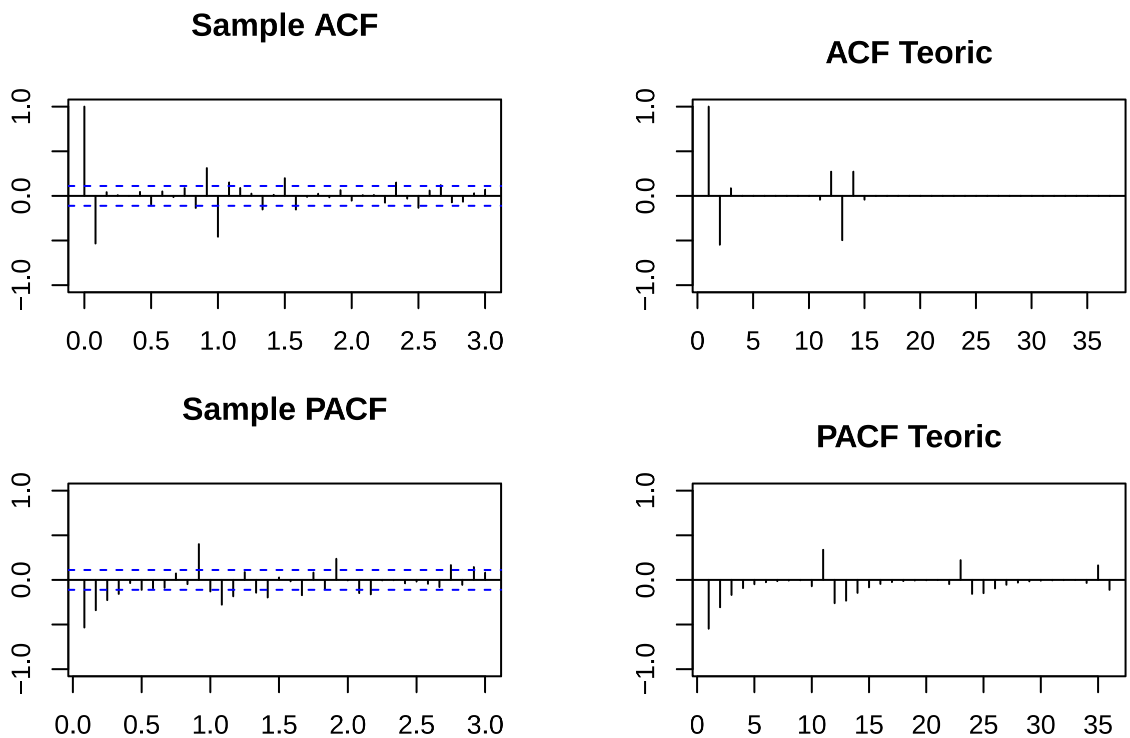

Series resid
Series resid
Series resid
1.0
0.5
0.0
−0.5
−1.0
0
1
2
3
4
5

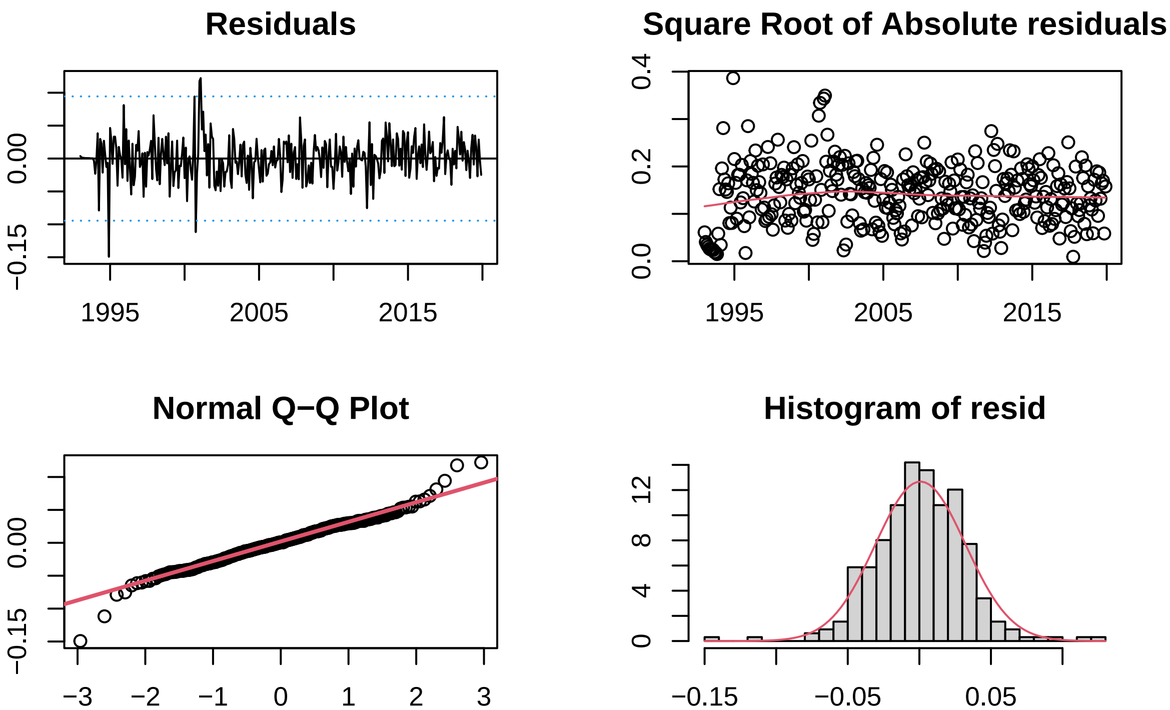



```
##
## ------------------------------------------------------------------
##
## Call:
## arima(x = lnserie, order = c(0, 1, 2), seasonal = list(order = c(0, 1, 1), period = 12))
##
## Coefficients:
##           ma1     ma2     sma1
##       -0.7872  0.1387  -0.8759
## s.e.   0.0595  0.0556   0.0451
##
## sigma^2 estimated as 0.001065:  log likelihood = 613.95,  aic = -1219.89
##
## Modul of AR Characteristic polynomial Roots:
##
## Modul of MA Characteristic polynomial Roots:  1.0111 1.0111 1.0111 1.0111 1.0111 1.0111 1.0111 1.011
##
## Psi-weights (MA(inf))
##
## --------------------
##      psi 1      psi 2      psi 3      psi 4      psi 5      psi 6      psi 7
## -0.7872180  0.1386801  0.0000000  0.0000000  0.0000000  0.0000000  0.0000000
##      psi 8      psi 9     psi 10     psi 11     psi 12     psi 13     psi 14
##  0.0000000  0.0000000  0.0000000  0.0000000 -0.8759337  0.6895508 -0.1214746
##     psi 15     psi 16     psi 17     psi 18     psi 19     psi 20
##  0.0000000  0.0000000  0.0000000  0.0000000  0.0000000  0.0000000
##
## Pi-weights (AR(inf))
##
```

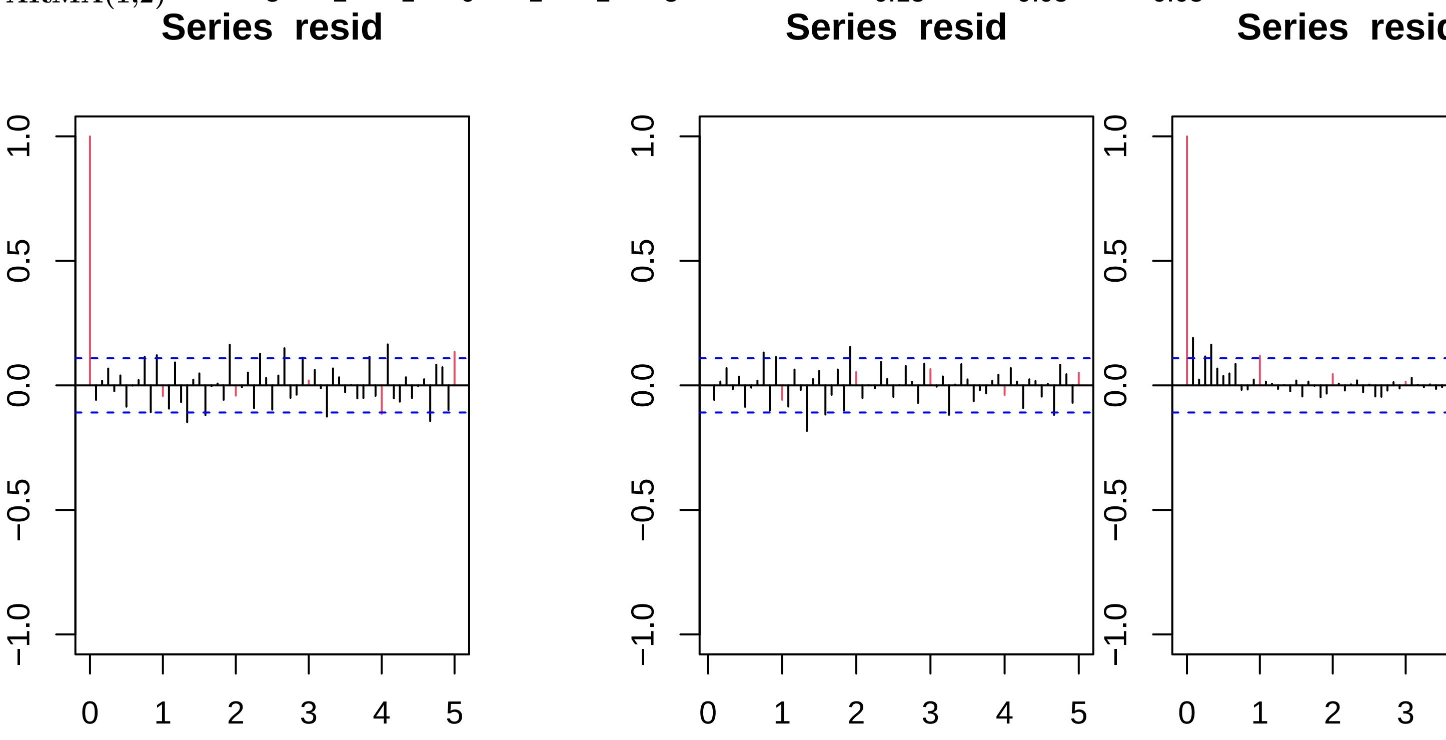
```
## --------------------
##         pi 1         pi 2         pi 3         pi 4         pi 5         pi 6
## -0.787218034 -0.481032141 -0.269505707 -0.145450171 -0.077125922 -0.040543873
##         pi 7         pi 8         pi 9        pi 10        pi 11        pi 12
## -0.021221038 -0.011082956 -0.005781767 -0.003014526 -0.001571273 -0.876752591
##        pi 13        pi 14        pi 15        pi 16        pi 17        pi 18
## -0.689977547 -0.421574638 -0.236185008 -0.127465088 -0.067588657 -0.035530140
##        pi 19        pi 20
## -0.018596766 -0.009712386
##
## Normality Tests
##
## --------------------
##
##  Shapiro-Wilk normality test
##
## data:  resid(model)
## W = 0.97599, p-value = 3.076e-05
##
##
##  Anderson-Darling normality test
##
## data:  resid(model)
## A = 0.73796, p-value = 0.05399
##
##  Jarque Bera Test
##
## data:  resid(model)
## X-squared = 71.851, df = 2, p-value = 2.22e-16
##
##
## Homoscedasticity Test
##
## --------------------
##
##  studentized Breusch-Pagan test
##
## data:  resid(model) ~ I(obs - resid(model))
## BP = 2.3425, df = 1, p-value = 0.1259
##
##
## Independence Tests
##
## --------------------
##
##  Durbin-Watson test
##
## data:  resid(model) ~ I(1:length(resid(model)))
## DW = 2.0239, p-value = 0.5635
## alternative hypothesis: true autocorrelation is greater than 0
##
##
## Ljung-Box test
```

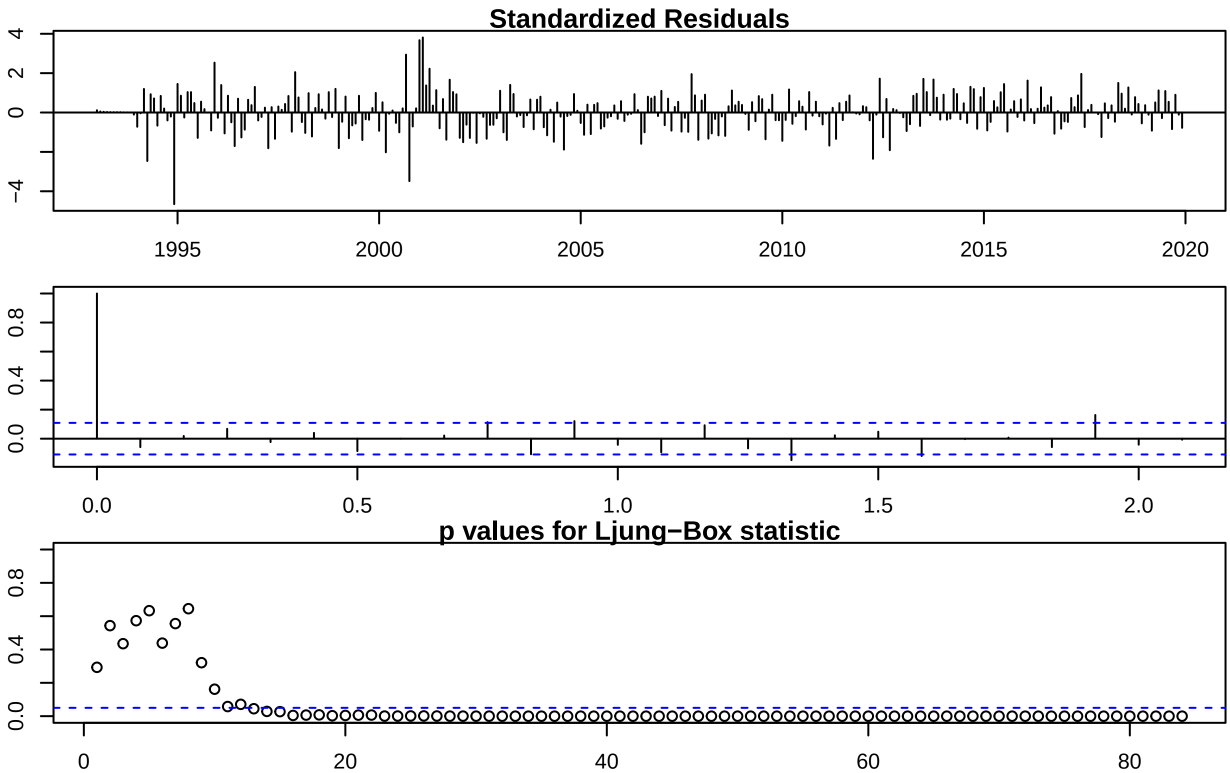
```
##      lag.df    statistic      p.value
## [1,]      1 1.830573e-04 9.892051e-01
## [2,]      2 4.321965e-01 8.056561e-01
## [3,]      3 9.062973e-01 8.239079e-01
## [4,]      4 9.863685e-01 9.118559e-01
## [5,]     12 2.149488e+01 4.358670e-02
## [6,]     24 5.472970e+01 3.403871e-04
## [7,]     36 8.541745e+01 6.762500e-06
## [8,]     48 1.098332e+02 9.496701e-07
```




Residuals
Square Root of Absolute residuals
Normal Q−Q Plot
Histogram of resid

**ARMA(1,2)**

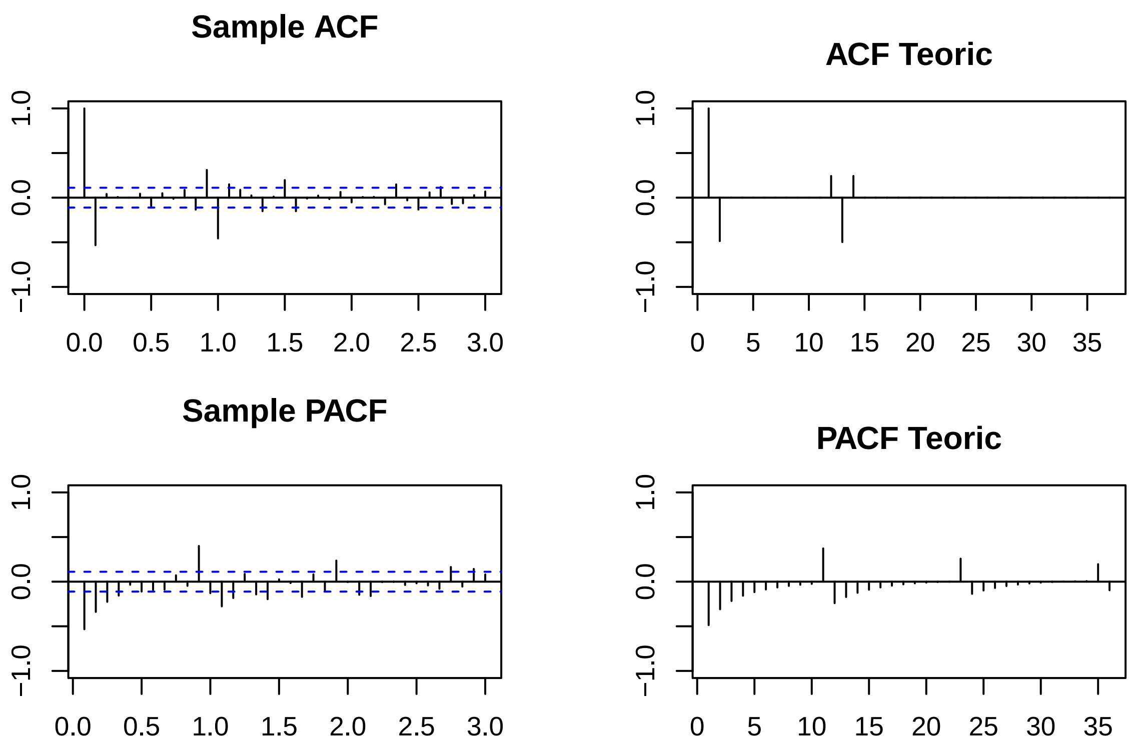

Series resid
Series resid



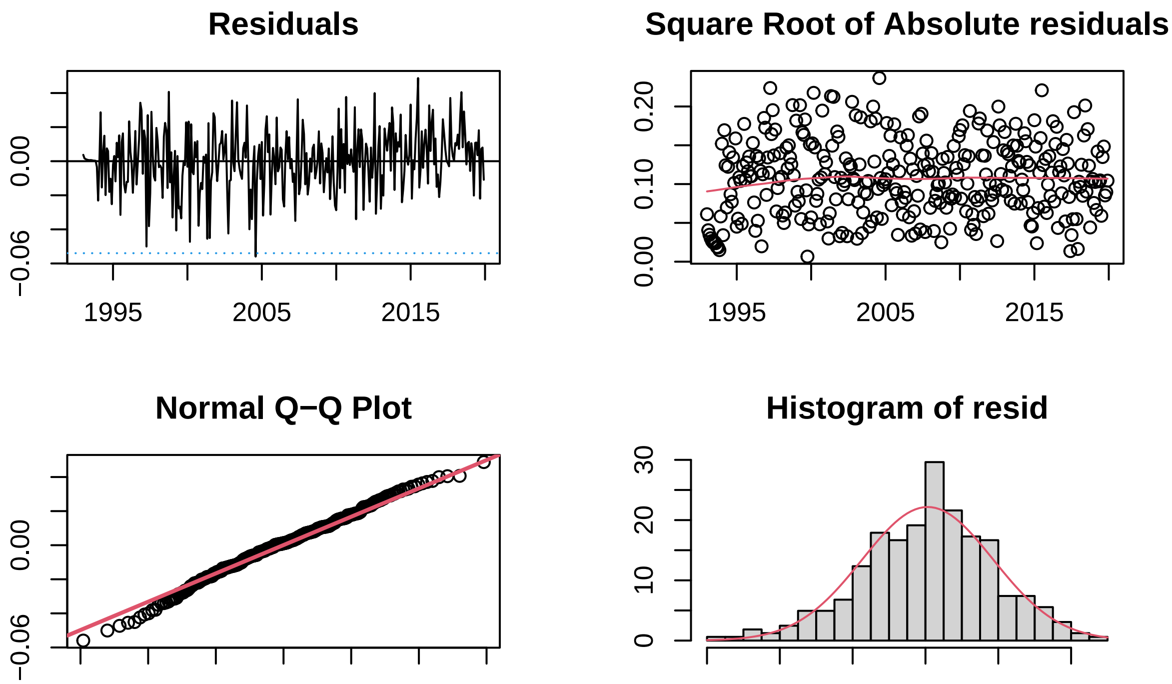

```
##
## --------------------------------------------------------------------
##
## Call:
## arima(x = lnserie, order = c(1, 1, 2), seasonal = list(order = c(0, 1, 1), period = 12))
##
## Coefficients:
##          ar1      ma1     ma2     sma1
##       0.9888  -1.7281  0.7451  -0.9219
## s.e.  0.0159   0.0377  0.0356   0.0452
##
## sigma^2 estimated as 0.001028:  log likelihood = 617.75,  aic = -1225.5
##
## Modul of AR Characteristic polynomial Roots:  1.011333
##
## Modul of MA Characteristic polynomial Roots:  1.006796 1.006796 1.006796 1.006796 1.006796 1.006796
##
## Psi-weights (MA(inf))
##
## --------------------
##         psi 1         psi 2         psi 3         psi 4         psi 5
## -0.7393437566  0.0140457644  0.0138883609  0.0137327214  0.0135788260
##         psi 6         psi 7         psi 8         psi 9        psi 10
##  0.0134266553  0.0132761898  0.0131274106  0.0129802986  0.0128348352
##        psi 11        psi 12        psi 13        psi 14        psi 15
##  0.0126910020 -0.9093917550  0.6940391319 -0.0006802582 -0.0006726349
##        psi 16        psi 17        psi 18        psi 19        psi 20
## -0.0006650970 -0.0006576436 -0.0006502738 -0.0006429865 -0.0006357809
##
```

```
## Pi-weights (AR(inf))
##
## --------------------
##         pi 1         pi 2         pi 3         pi 4         pi 5         pi 6
## -0.739343757 -0.532583426 -0.369489222 -0.241698013 -0.142379420 -0.065961007
##         pi 7         pi 8         pi 9        pi 10        pi 11        pi 12
## -0.007902188  0.035491750  0.067222569  0.089724780  0.104968927 -0.807394119
##        pi 13        pi 14        pi 15        pi 16        pi 17        pi 18
## -0.561891823 -0.369433552 -0.219764002 -0.104515917 -0.016870797  0.048720184
##        pi 19        pi 20
##  0.096765665  0.130922746
##
## Normality Tests
##
## --------------------
##
##  Shapiro-Wilk normality test
##
## data:  resid(model)
## W = 0.9754, p-value = 2.419e-05
##
##
##  Anderson-Darling normality test
##
## data:  resid(model)
## A = 0.81635, p-value = 0.03455
##
##
##  Jarque Bera Test
##
## data:  resid(model)
## X-squared = 73.466, df = 2, p-value < 2.2e-16
##
##
## Homoscedasticity Test
##
## --------------------
##
##  studentized Breusch-Pagan test
##
## data:  resid(model) ~ I(obs - resid(model))
## BP = 3.1928, df = 1, p-value = 0.07397
##
##
## Independence Tests
##
## --------------------
##
##  Durbin-Watson test
##
## data:  resid(model) ~ I(1:length(resid(model)))
## DW = 2.1273, p-value = 0.8629
## alternative hypothesis: true autocorrelation is greater than 0
##
```

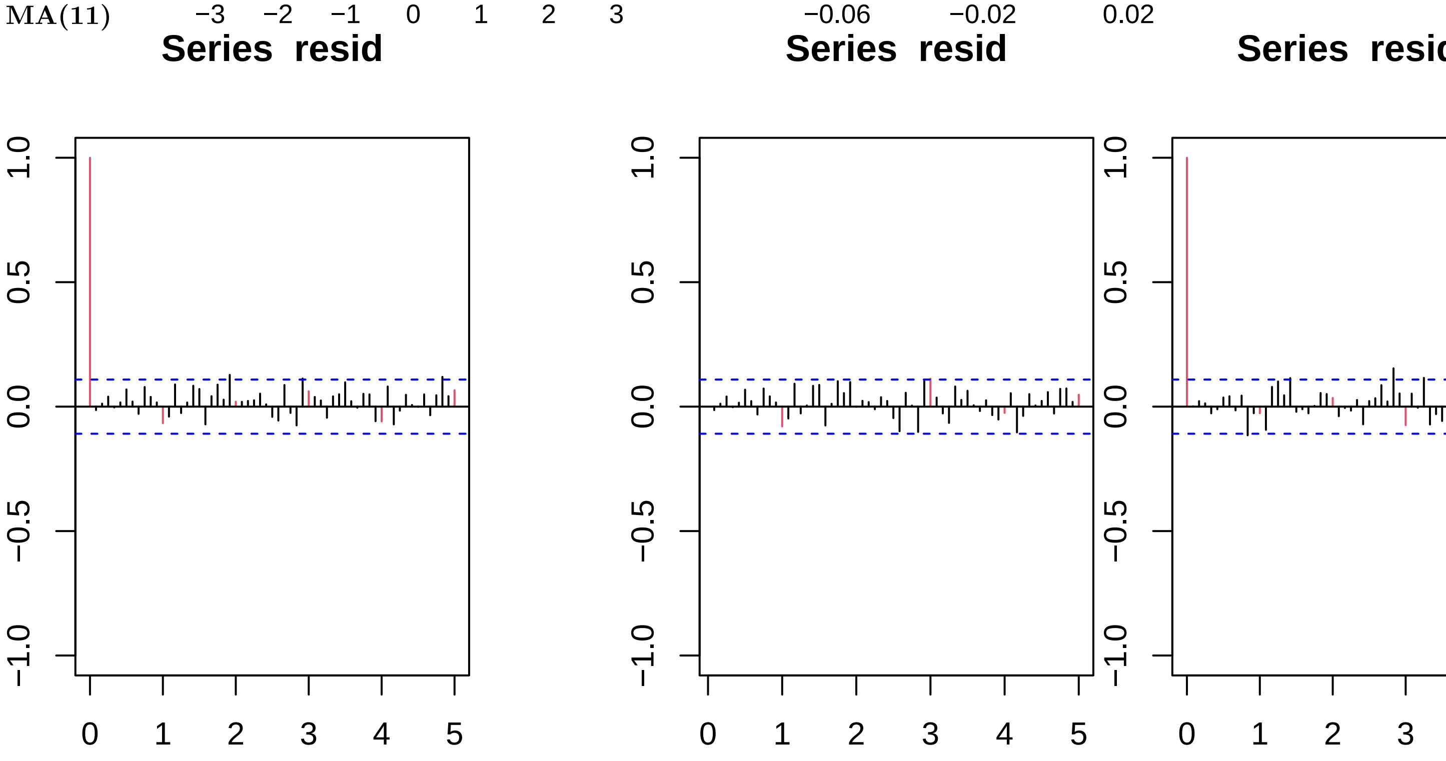
```
##
## Ljung-Box test
##      lag.df  statistic      p.value
## [1,]       1   1.105301 2.931057e-01
## [2,]       2   1.220740 5.431498e-01
## [3,]       3   2.730671 4.350403e-01
## [4,]       4   2.910458 5.729202e-01
## [5,]      12  19.796406 7.103663e-02
## [6,]      24  51.747430 8.439298e-04
## [7,]      36  79.893651 3.581260e-05
## [8,]      48 102.282066 8.457058e-06
```

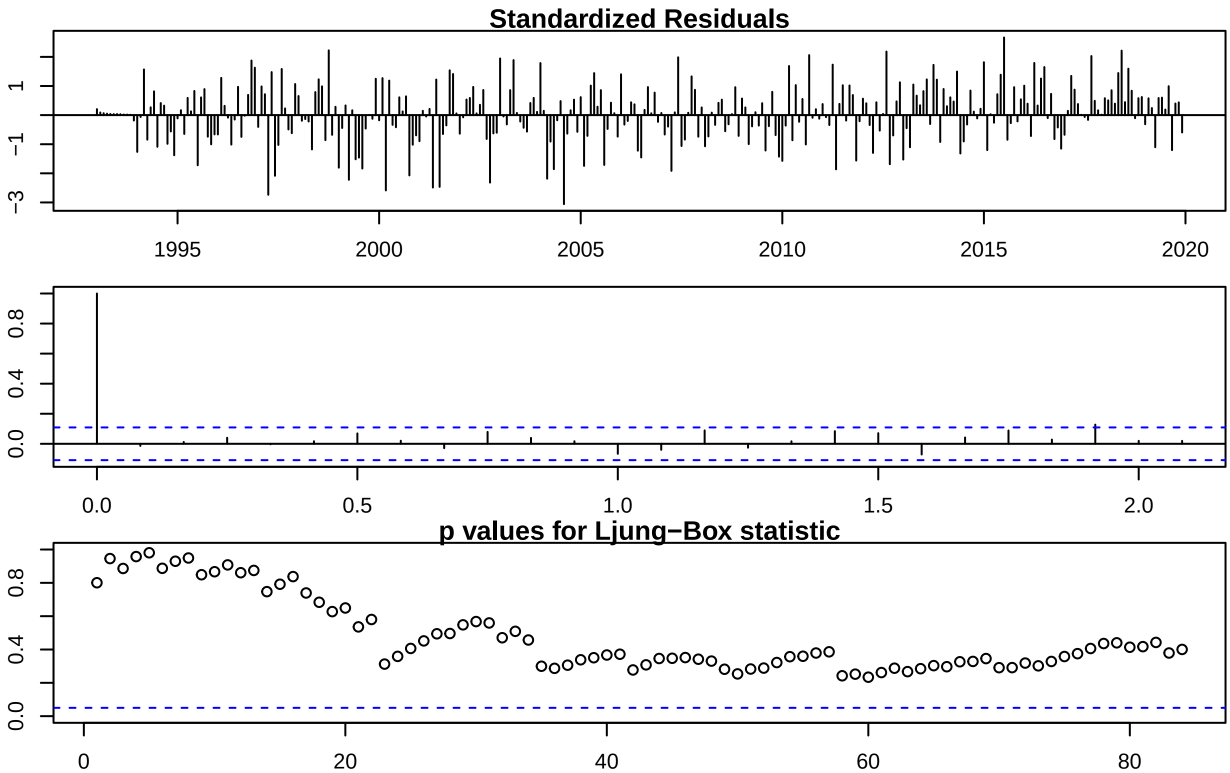



**ARIMAX models validation detailed**


Residuals
Square Root of Absolute residuals
Normal Q−Q Plot
Histogram of resid

**MA(11)**

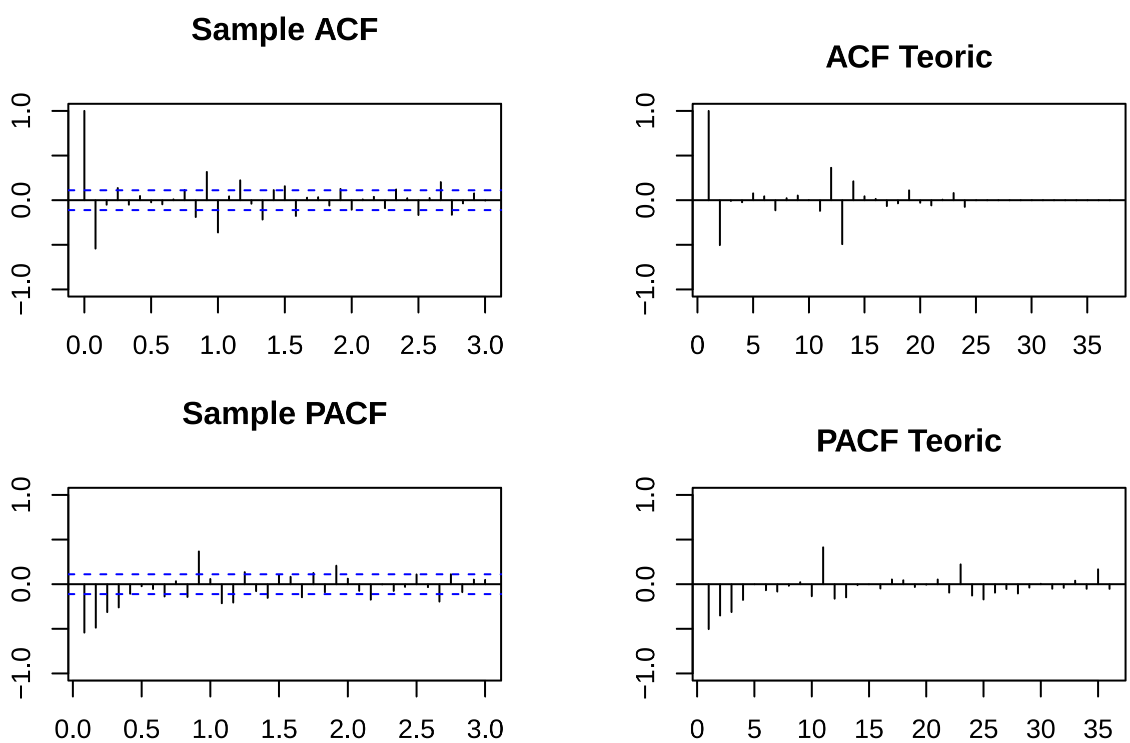

Series resid
Series resid

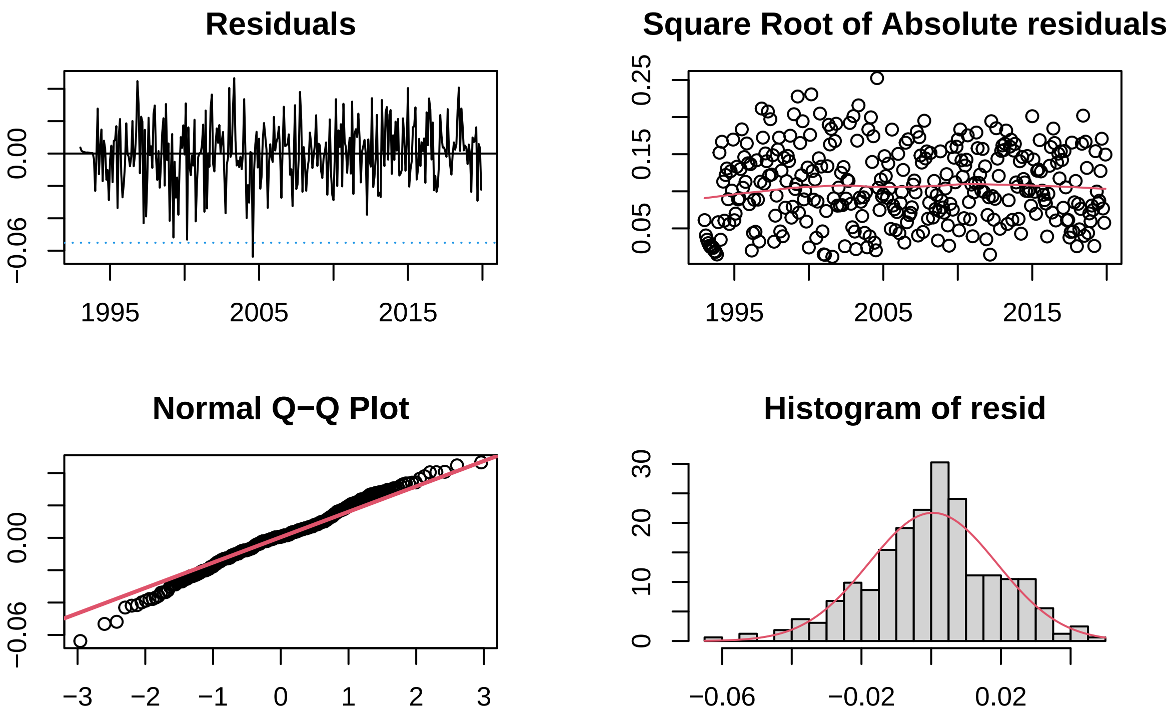



```
##
## --------------------------------------------------------------------
##
## Call:
## arima(x = lnserie.lin, order = c(0, 1, 11), seasonal = list(order = c(0, 1,
##     1), period = 12), xreg = data.frame(wTradDays, wEast))
##
## Coefficients:
##           ma1      ma2      ma3     ma4     ma5      ma6     ma7     ma8
##       -0.7307  -0.0350  -0.0120  0.1591  0.0268  -0.2606  0.1643  0.1506
## s.e.   0.0579   0.0707   0.0694  0.0738  0.0730   0.0846  0.0698  0.0694
##           ma9     ma10    ma11     sma1  wTradDays   wEast
##       -0.0780  -0.0910  0.2677  -0.8355      4e-03  0.0274
## s.e.   0.0773   0.0705  0.0682   0.0447      3e-04  0.0050
##
## sigma^2 estimated as 0.0003348:  log likelihood = 795.44,  aic = -1560.89
##
## Modul of AR Characteristic polynomial Roots:
##
## Modul of MA Characteristic polynomial Roots:  1.015086 1.015086 1.015086 1.015086 1.015086 1.015086
##
## Psi-weights (MA(inf))
##
## --------------------
##        psi 1        psi 2        psi 3        psi 4        psi 5        psi 6
## -0.730663841 -0.035034471 -0.011958593  0.159102034  0.026766312 -0.260585532
##        psi 7        psi 8        psi 9       psi 10       psi 11       psi 12
##  0.164289738  0.150622696 -0.077974686 -0.090955133  0.267678790 -0.835533496
##       psi 13       psi 14       psi 15       psi 16       psi 17       psi 18
```

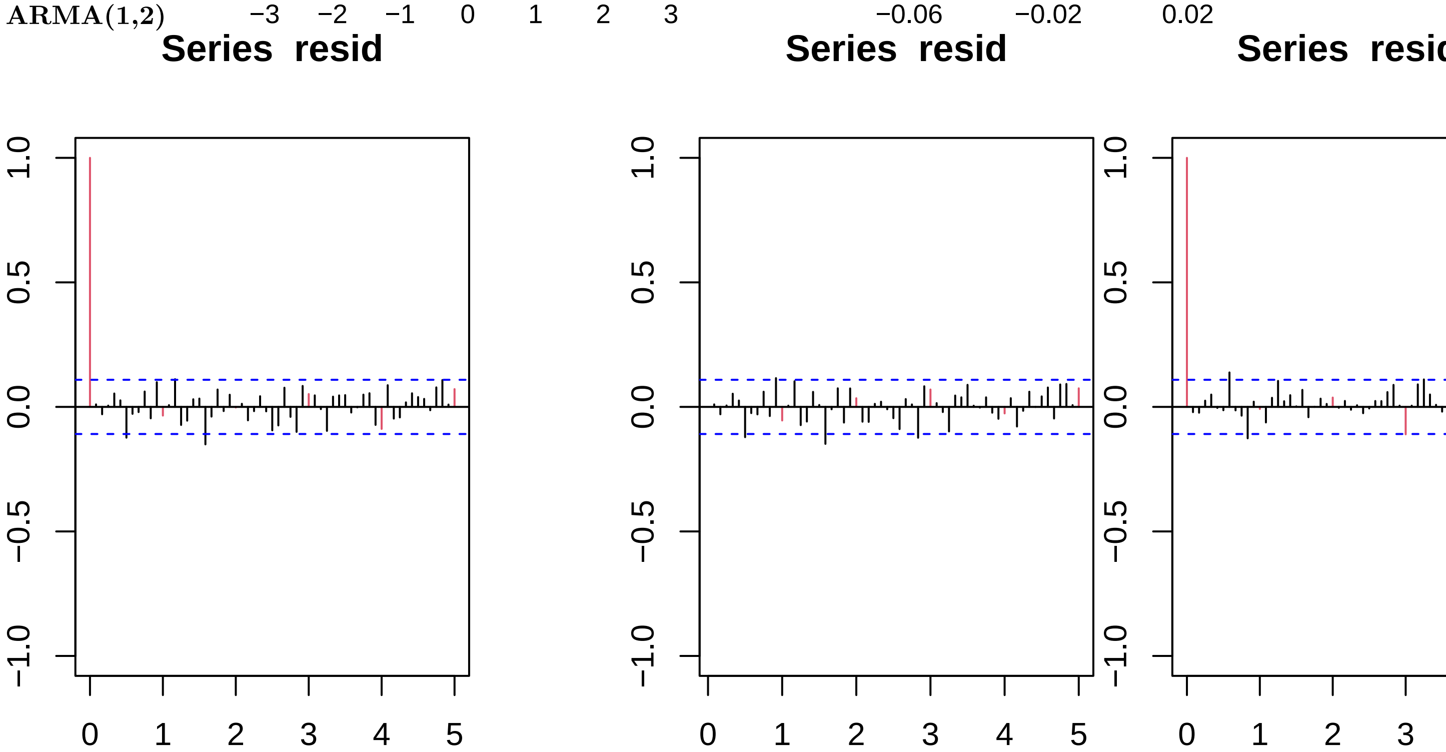
```
##  0.610494114  0.029272474  0.009991805 -0.132935079 -0.022364151  0.217727940
##       psi 19       psi 20
## -0.137269580 -0.125850308
##
## Pi-weights (AR(inf))
##
## --------------------
##         pi 1         pi 2         pi 3         pi 4         pi 5         pi 6
## -0.730663841 -0.568904120 -0.453234684 -0.200729128 -0.026331230 -0.182206318
##         pi 7         pi 8         pi 9        pi 10        pi 11        pi 12
## -0.075227168  0.104818760  0.088773513  0.057246598  0.314282095 -0.521917433
##        pi 13        pi 14        pi 15        pi 16        pi 17        pi 18
## -0.276875206 -0.060185112 -0.057080006 -0.003345065  0.162082310 -0.025053087
##        pi 19        pi 20
## -0.051236018  0.080051407
##
## Normality Tests
##
## --------------------
##
##  Shapiro-Wilk normality test
##
## data:  resid(model)
## W = 0.99367, p-value = 0.1933
##
##
##  Anderson-Darling normality test
##
## data:  resid(model)
## A = 0.53764, p-value = 0.1671
##
##
##  Jarque Bera Test
##
## data:  resid(model)
## X-squared = 3.695, df = 2, p-value = 0.1576
##
##
## Homoscedasticity Test
##
## --------------------
##
##  studentized Breusch-Pagan test
##
## data:  resid(model) ~ I(obs - resid(model))
## BP = 2.7103, df = 1, p-value = 0.0997
##
##
## Independence Tests
##
## --------------------
##
##  Durbin-Watson test
##
```

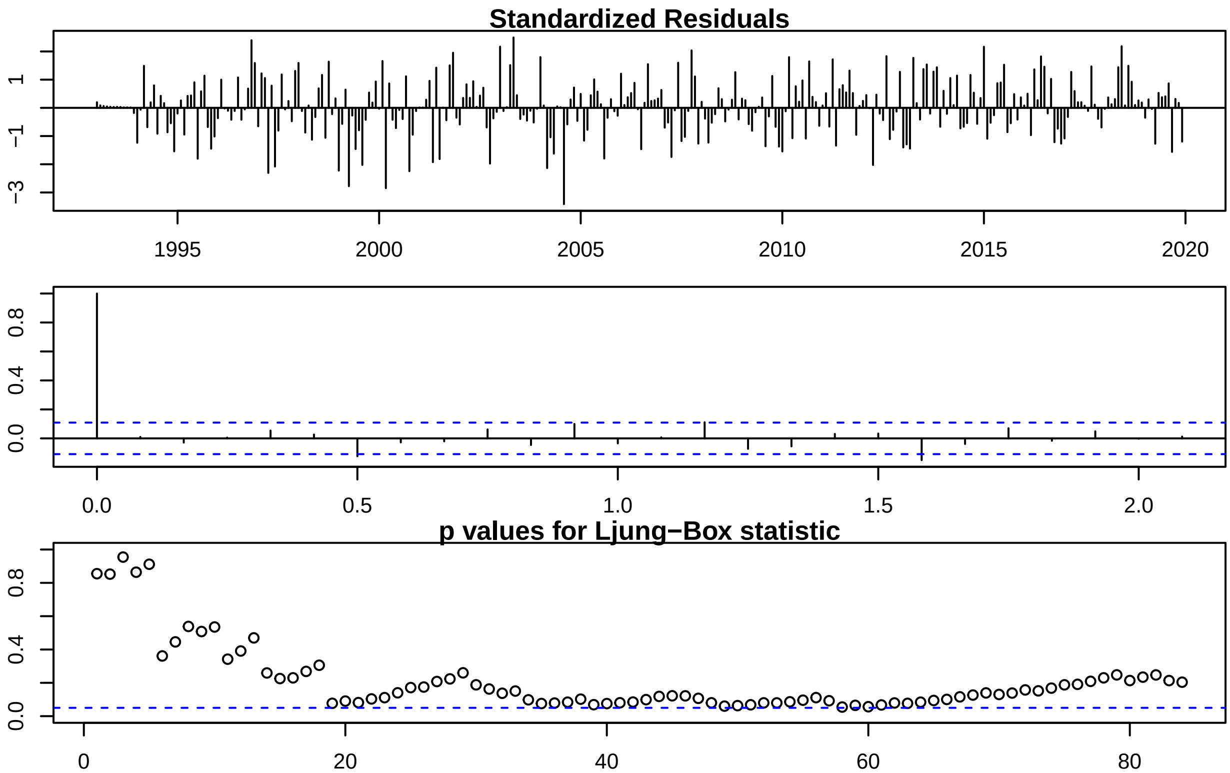
```
## data:  resid(model) ~ I(1:length(resid(model)))
## DW = 2.0794, p-value = 0.7457
## alternative hypothesis: true autocorrelation is greater than 0
##
##
## Ljung-Box test
##       lag.df   statistic   p.value
## [1,]       1  0.06374809 0.8006671
## [2,]       2  0.11183568 0.9456168
## [3,]       3  0.64387535 0.8863187
## [4,]       4  0.64728808 0.9576667
## [5,]      12  6.95104385 0.8608317
## [6,]      24 25.89071009 0.3587337
## [7,]      36 40.26648668 0.2870097
## [8,]      48 51.72522820 0.3305222
```




Residuals
Square Root of Absolute residuals
Normal Q−Q Plot
Histogram of resid


**ARMA(1,2)**

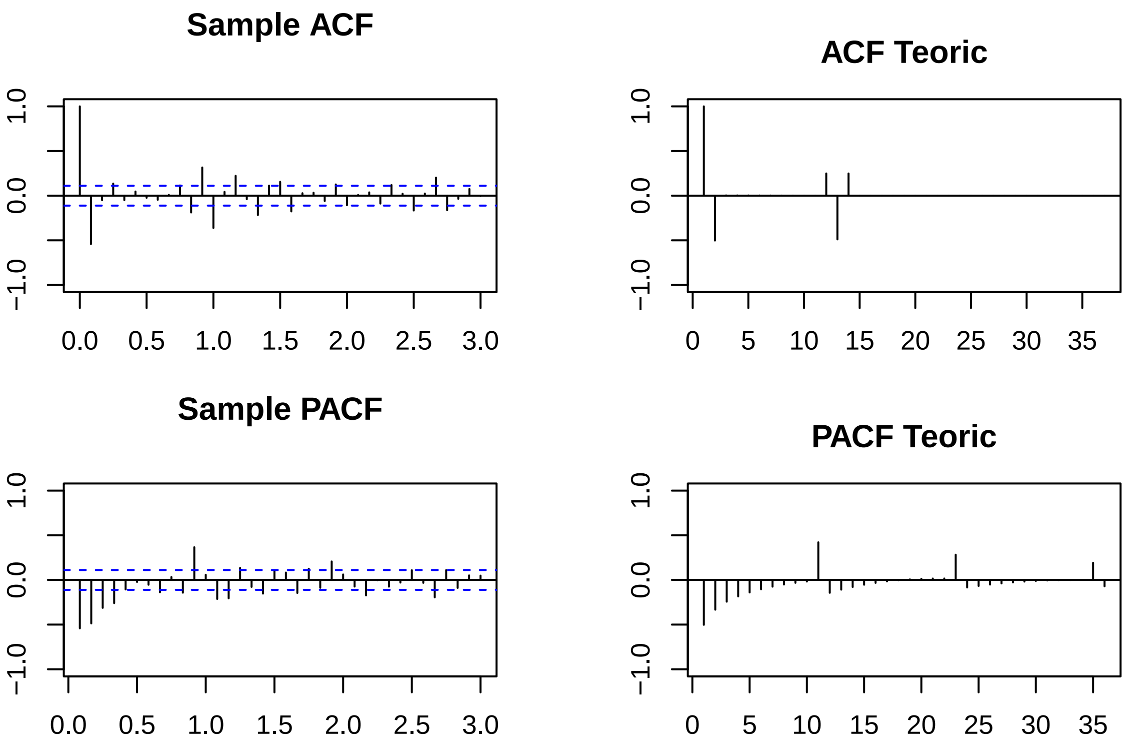

Series resid
Series resid




```
##
## ------------------------------------------------------------------
##
## Call:
## arima(x = lnserie.lin, order = c(1, 1, 2), seasonal = list(order = c(0, 1, 1),
##     period = 12), xreg = data.frame(wTradDays, wEast))
##
## Coefficients:
##          ar1      ma1     ma2     sma1  wTradDays   wEast
##       0.9703  -1.7674  0.8044  -0.8272     0.0042  0.0334
## s.e.  0.0231   0.0363  0.0368   0.0413     0.0003  0.0051
##
## sigma^2 estimated as 0.0003489:  log likelihood = 789.87,  aic = -1565.74
##
## Modul of AR Characteristic polynomial Roots:  1.030661
##
## Modul of MA Characteristic polynomial Roots:  1.015937 1.015937 1.015937 1.015937 1.015937 1.015937
##
## Psi-weights (MA(inf))
##
## --------------------
##        psi 1        psi 2        psi 3        psi 4        psi 5        psi 6
## -0.797110609  0.030960150  0.030039126  0.029145501  0.028278461  0.027437213
##        psi 7        psi 8        psi 9       psi 10       psi 11       psi 12
##  0.026620992  0.025829052  0.025060671  0.024315149  0.023591805 -0.804285868
##       psi 13       psi 14       psi 15       psi 16       psi 17       psi 18
##  0.681559676 -0.004061146 -0.003940332 -0.003823112 -0.003709380 -0.003599030
##       psi 19       psi 20
## -0.003491964 -0.003388083
```

```
##
## Pi-weights (AR(inf))
##
## --------------------
##        pi 1        pi 2        pi 3        pi 4        pi 5        pi 6
## -0.79711061 -0.60442517 -0.42707593 -0.26862365 -0.13123338 -0.01586735
##        pi 7        pi 8        pi 9       pi 10       pi 11       pi 12
##  0.07751523  0.14976050  0.20233102  0.23713112  0.25634998 -0.56485091
##       pi 13       pi 14       pi 15       pi 16       pi 17       pi 18
## -0.40192463 -0.25600408 -0.12916066 -0.02235477  0.06438242  0.13176826
##       pi 19       pi 20
##  0.18109571  0.21407284
##
## Normality Tests
##
## --------------------
##
##  Shapiro-Wilk normality test
##
## data:  resid(model)
## W = 0.99203, p-value = 0.0795
##
##
##  Anderson-Darling normality test
##
## data:  resid(model)
## A = 0.81168, p-value = 0.03548
##
##
##  Jarque Bera Test
##
## data:  resid(model)
## X-squared = 3.7374, df = 2, p-value = 0.1543
##
##
## Homoscedasticity Test
##
## --------------------
##
##  studentized Breusch-Pagan test
##
## data:  resid(model) ~ I(obs - resid(model))
## BP = 1.9448, df = 1, p-value = 0.1632
##
##
## Independence Tests
##
## --------------------
##
##  Durbin-Watson test
##
## data:  resid(model) ~ I(1:length(resid(model)))
## DW = 1.9933, p-value = 0.4537
## alternative hypothesis: true autocorrelation is greater than 0
```

```
##
##
## Ljung-Box test
##      lag.df   statistic    p.value
## [1,]      1  0.03339708 0.85499521
## [2,]      2  0.31951835 0.85234903
## [3,]      3  0.32805937 0.95466771
## [4,]      4  1.28477100 0.86395401
## [5,]     12 12.70239030 0.39103817
## [6,]     24 31.49435268 0.14003439
## [7,]     36 48.57482633 0.07862066
## [8,]     48 62.32401136 0.08014423
```




**Models outputs**

**SARIMA(1,1,2)x(0,1,1)12**

```
##
## Call:
## arima(x = lnserie, order = c(1, 1, 2), seasonal = list(order = c(0, 1, 1), period = 12))
##
## Coefficients:
##          ar1      ma1     ma2     sma1
##       0.9888  -1.7281  0.7451  -0.9219
## s.e.  0.0159   0.0377  0.0356   0.0452
##
## sigma^2 estimated as 0.001028:  log likelihood = 617.75,  aic = -1225.5
```

**SARIMAX(1,1,2)x(0,1,1)12 with outlier and calendar effects treatment**

```
##
## Call:
## arima(x = lnserie.lin, order = c(1, 1, 2), seasonal = list(order = c(0, 1, 1),
##     period = 12), xreg = data.frame(wTradDays, wEast))
##
## Coefficients:
##          ar1      ma1     ma2     sma1  wTradDays   wEast
##       0.9703  -1.7674  0.8044  -0.8272     0.0042  0.0334
## s.e.  0.0231   0.0363  0.0368   0.0413     0.0003  0.0051
##
## sigma^2 estimated as 0.0003489:  log likelihood = 789.87,  aic = -1565.74
```